Design, Modelling and Simulation of a Dual-Stage EV Fast-Charging System with Automatic Parameter Calculation

Active-Front-End Rectifier with LCL Filter and Synchronous Buck-Boost Battery Charger (up to 200 kW)

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Abstract

This report presents the design and MATLAB/Simulink simulation of a dual-stage electric-vehicle charging system with a rated power of up to 200 kW. The system consists of a three-phase Active Front End (AFE) rectifier with an LCL filter, connected to a 50 Hz grid, followed by a synchronous buck-boost DC/DC converter that charges a 100 kWh battery at up to 2C (500 A). A MATLAB script takes the grid line-to-line voltage, the required charging power, the initial battery state of charge (SOC) and the simulation time as inputs. It calculates the DC-link voltage, the DC-link capacitor, the LCL filter, the buck-boost inductor and capacitor, the duty-cycle limit and all controller gains, prints each calculation in the Command Window, runs the Simulink model, and reports the power, efficiency and power factor. Four cases were simulated: three at 400 V and 200 kW (5 s at 80 % SOC, 10 s at 70 % and 3 s at 90 %) and one at 220 V and 60 kW (0.5 s at 70 %). In every case the DC link was regulated, the battery was charged first at constant current (CC) and then at constant voltage (CV), and the SOC increased. The grid-side displacement power factor was 1.000, and the overall efficiency in the CC stage was about 90 % at 400 V and about 93 % at 220 V.

1. Introduction

1.1 General introduction to EV charging

The number of electric vehicles (EVs) is growing quickly, and with it the need for charging infrastructure that is fast, reliable and compatible with the grid [7], [9]. Charging is commonly grouped into Level 1 (convenience), Level 2 (primary) and Level 3 (fast) charging, and chargers are either on-board, where power is limited by weight, space and cost, or off-board, which can be designed for high charging rates and is less constrained by size and weight [9].

A DC fast charger takes power from the AC grid and delivers a controlled DC current and voltage to the battery. The usual structure has two conversion stages: an AC–DC stage, ideally an active front-end (AFE) rectifier that draws sinusoidal current at unity power factor and regulates a DC link, followed by a DC–DC stage that controls the battery charging current and voltage [8]. The battery is normally charged in constant-current (CC) mode and then in constant-voltage (CV) mode.

1.2 Literature survey

Charging infrastructure. Yilmaz and Krein [9] review battery chargers, charging power levels and infrastructure for plug-in electric and hybrid vehicles, classifying chargers as on-board or off-board with unidirectional or bidirectional power flow. Tu et al. [7] concentrate on extreme fast charging, presenting the design considerations of such stations, the typical power-electronic converter topologies and the use of solid-state transformers. Franzese et al. [8] give an overview of ultra-fast charging stations, covering their effect on the distribution grid, four active front-end AC–DC converters, three isolated DC–DC converters and two control architectures, together with the main standards. Sharida et al. [14] review fast-charger topologies, protection, communication and control, and the interaction between fast chargers and the grid.

Grid-side converter and its control. Kazmierkowski and Malesani [10] survey the current-control techniques for three-phase voltage-source PWM converters, including hysteresis, linear PI controllers in rotating coordinates and predictive controllers. When an LCL filter is placed between the converter and the grid, Liserre et al. [4] present a design procedure for the filter and the control of the resulting active rectifier, and Dannehl et al. [11] analyse the limits within which voltage-oriented PI current control remains stable with an LCL filter. The DC-link capacitor of the rectifier can be sized with the method of Hava et al. [2]. A constant-current constant-voltage fast battery charger is described in [1], and the modelling and control of a voltage source converter is treated in [6].

DC–DC stage. Application notes such as [5] give the standard design equations for the buck-boost power stage, which are the basis of the inductor and capacitor calculations used here.

Battery model. Tremblay et al. [13] propose a generic dynamic battery model for vehicle simulation, and Tremblay and Dessaint [12] validate it experimentally for four battery types and include it in the SimPowerSystems simulation software. The Battery block of Simscape Electrical used in this work is based on this family of models.

Position of this work. The cited works describe topologies, standards, controllers and models separately. This report combines them in one MATLAB/Simulink study of a 200 kW two-stage charger in which the grid voltage, the required power and the initial battery state of charge are the inputs, and in which every component value and controller gain is calculated automatically and the power, efficiency and power factor are reported.

1.3 Objectives

- to design a dual-stage charger, with an AFE rectifier and LCL filter on a 50 Hz grid and a synchronous buck-boost converter feeding a 100 kWh (400 V, 250 Ah) battery limited to 2C (500 A), for a rated power of up to 200 kW;

- to automate the design: the user enters the grid voltage, the required power, the initial SOC and the simulation time, and all components and control gains are calculated and passed to the model;
- to verify in Simulink, for different grid voltages, powers and initial SOC values, that the battery is charged in CC/CV mode, that its SOC increases, and that the DC link, the power factor and the efficiency are acceptable.

1.4 System specifications

Table 1: System specifications and the inputs of the four cases.

Parameter	Value
Grid (line-to-line, RMS), 50 Hz	input: 400 V (Cases 1–3), 220 V (Case 4)
Charging power	input: 200 kW (Cases 1–3), 60 kW (Case 4); maximum 200 kW
Initial battery SOC	input: 80 %, 70 %, 90 %, 70 % (Cases 1–4)
Simulation time	input: 5 s, 10 s, 3 s, 0.5 s (Cases 1–4)
Battery	100 kWh, 400 V nominal, 250 Ah (2C = 500 A)
AFE switching frequency	10 kHz, with an LCL filter
DC/DC switching frequency	100 kHz, synchronous buck-boost
CV target voltage	447 V (1.1175 × 400 V)

2. Methodology

This section presents the circuit diagram of the simulated system, the charging profile, the design formulas and calculations, the component values, and the assumptions.

2.1 Circuit diagram (Simulink model)

Figure 1 shows the top level of the model (SYSTEM_MODEL_param_RAMP_SOC.slx). The grid feeds the AFE rectifier through the LCL filter; the rectifier output is the DC link, which feeds the synchronous buck-boost converter; the converter charges the battery. The Close loop control subsystem produces the AFE modulation reference, and the buck-boost duty cycle is produced by a PI current controller. The powergui is set to discrete simulation with a 3 µs sample time.

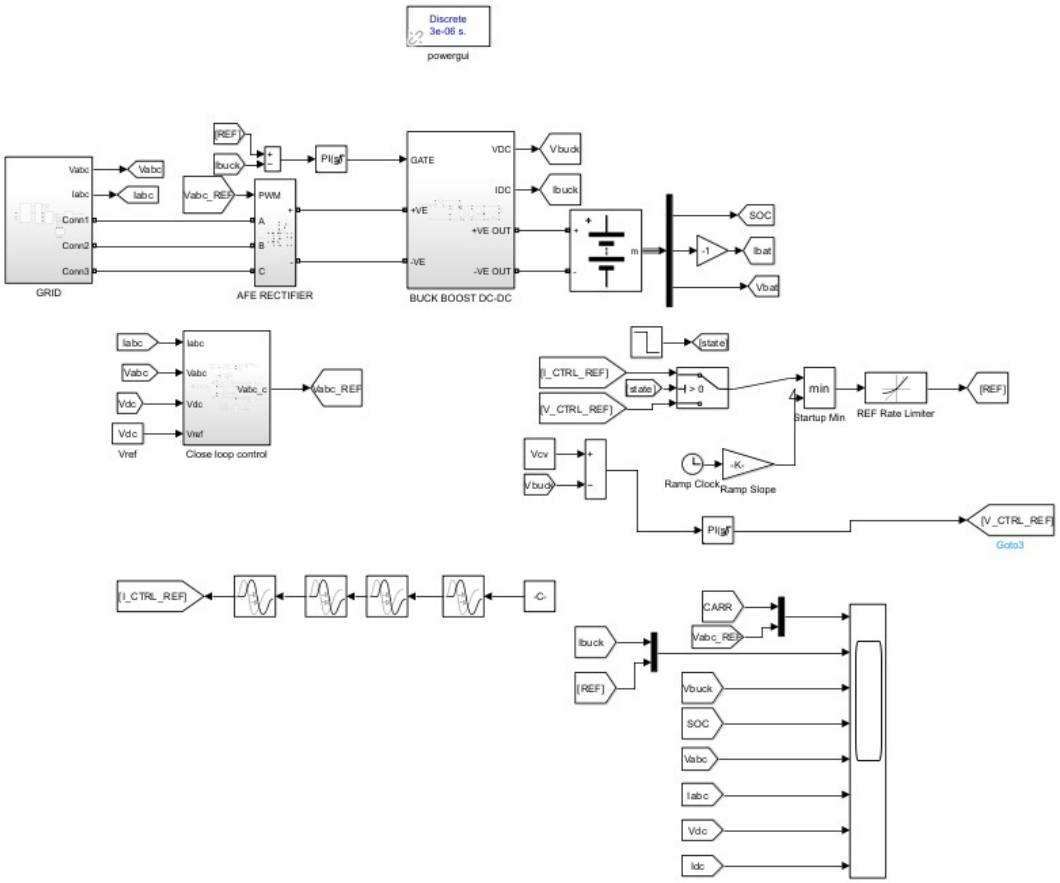


Figure 1: Top level of the Simulink model.

The grid subsystem (Figure 2) contains the programmable three-phase source, the voltage and current measurement, the converter-side branch (L, R), the grid-side branch (Lf, Rf) and the shunt filter-capacitor branch. All values come from workspace variables calculated by the script, and the source amplitude is set by V_grid.

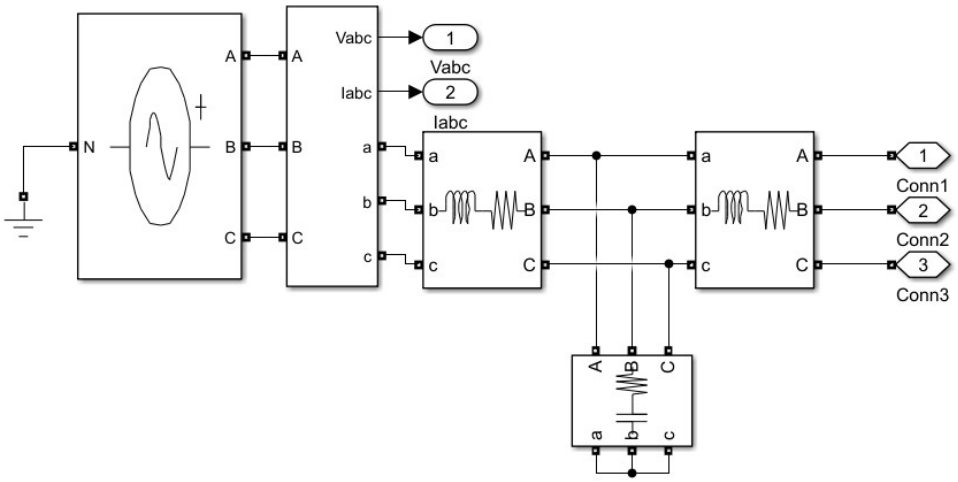


Figure 2: Grid subsystem with the LCL filter.

The AFE rectifier (Figure 3) is a three-leg IGBT/diode bridge with the DC-link capacitor Cin (value Cdc, initially charged to Vdc). The modulation reference is scaled by Kmod and compared with a 10 kHz triangular carrier to generate the six gate signals.

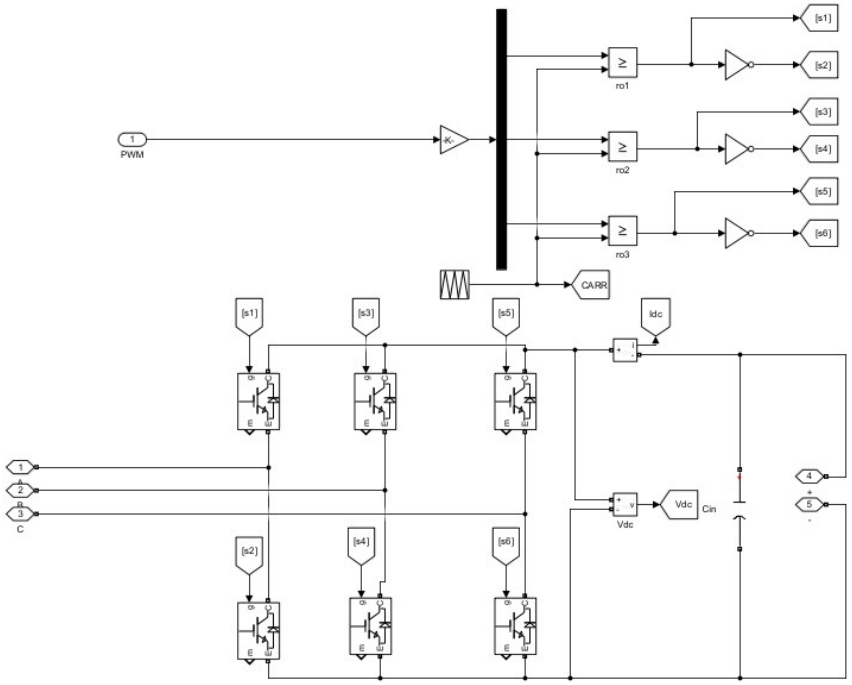


Figure 3: AFE rectifier subsystem.

The AFE control subsystem (Figure 4) implements voltage-oriented control. A PLL extracts the grid angle; the grid currents and voltages are transformed to the dq frame; the DC-link PI controller sets the d-axis current reference and the q-axis reference is zero; two PI current controllers with grid-voltage feed-forward and ωLt decoupling generate Vd and Vq, which are transformed back to abc.

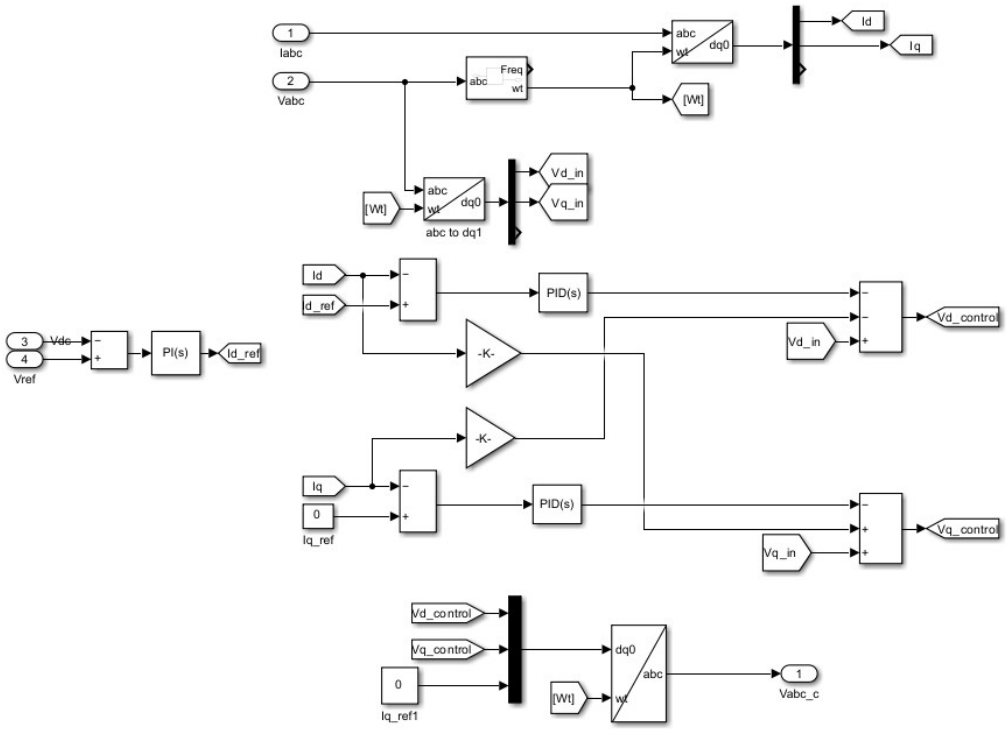


Figure 4: AFE control subsystem (voltage-oriented control).

The buck-boost subsystem (Figure 5) is an inverting synchronous buck-boost converter following [5]. The gate signal from the PI current controller drives the high-side IGBT through a PWM generator, and the second switch is driven by the complementary (NOT) signal. The inductor sits across the switching node, an output capacitor bank and a small series inductor form the output filter, and a voltage and a current measurement provide the feedback signals (Vbuck and Ibuck).

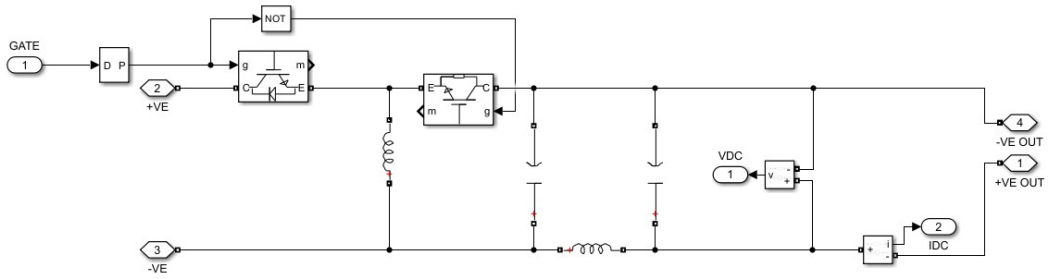


Figure 5: Synchronous buck-boost DC/DC converter subsystem.

The battery is the generic Battery block of Simscape Electrical, set to Lithium-Ion, 400 V nominal voltage and 250 Ah rated capacity (100 kWh). The initial SOC is the workspace variable SOC0, which the script sets from the user input (Figure 6 shows the evaluated value, 70 %). Temperature and aging effects are not simulated.

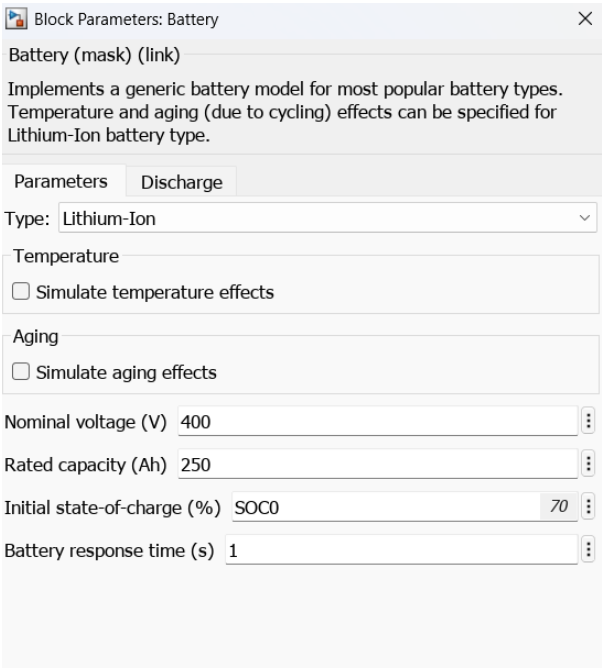


Figure 6: Battery block parameters with the initial SOC set by the variable SOC0.

2.2 Charging profile and start-up ramp

The DC/DC stage works in CC/CV mode. In CC mode the current reference follows a staircase of 40 %, 60 %, 80 % and 100 % of Io,max (200, 300, 400 and 500 A for 200 kW) with 0.07 s between steps, produced by a chain of transport delays. At t = 0.4 s a control signal switches the converter to CV mode, in which an outer PI voltage loop holds the charging voltage at 447 V and produces the current reference. The switch is time-based; it does not depend on the battery voltage. The reference passes through two blocks added for smoother operation (Figure 1): a start-up ramp (clock, gain and minimum block) that makes the reference rise from 0 A, and a rate limiter that limits the rise and fall of the reference to 5 kA/s (scaled with Io,max). They limit the DC-link disturbance caused by sudden load steps.

2.3 Design workflow and automatic calculation

The script charger_design_RAMP_v3.m (Appendix A) asks for the grid line-to-line RMS voltage, the required power, the initial SOC and the simulation time. It evaluates the closed-form design equations of Sections 2.4 to 2.7, prints every calculated variable with its unit and formula in the Command Window together with the design checks, and writes every value the Simulink model needs (V_grid, SOC0, Vdc, Cdc, L, Lf, Cf, R, Rf, Lt, Fsw, kpi, kii, kpv, kiv, Kmod, Io_max, Lch, Co, Dlim, kpb, kib, Vcv) into the base workspace. It then runs the model with logging blocks and calculates the power, efficiency and power factor (Section 2.8). The model uses these variables instead of hard-coded numbers. No toolbox other than Simulink and Simscape Electrical is required.

2.4 AFE rectifier and DC link

The DC-link voltage must exceed the peak of the diode-rectified line voltage. A 20 % margin is added and the result is rounded up to the next 50 V. The DC-link capacitor is selected for a 10 % voltage ripple:

$$Vdc,min = \sqrt{2} \cdot V(L\text{-}L) \qquad Vdc = ceil(1.2 \cdot Vdc,min \,/\, 50) \cdot 50$$

$$Cdc = P \,/\, (240 \cdot \Delta Vdc \cdot Vph \cdot f_grid), \quad \Delta Vdc = 0.1 \cdot Vdc, \quad Vph = V(L\text{-}L) \,/\, \sqrt{3}$$

2.5 LCL filter

The LCL filter follows the method of Liserre et al. [4] with a 20 % converter-side current ripple, 5 % attenuation and a filter capacitor of 5 % of the base capacitance:

$$Zb = V^2 / P \qquad Cb = 1 / (\omega s \cdot Zb) \qquad Iripple = 0.2 \cdot \sqrt{2} \cdot P / (\sqrt{3} \cdot V)$$

$$L = Vdc / (8 \cdot Fsw \cdot Iripple) \qquad Cf = 0.05 \cdot Cb \qquad Lf = r \cdot L \qquad Lt = L + Lf$$

$$Fres = (1/2\pi) \cdot \sqrt{(Lt / (L \cdot Lf \cdot Cf))} \qquad 6Fs < Fres < 0.5Fsw$$

2.6 AFE controller

Voltage-oriented control (VOC) in the dq frame is used. The outer DC-link voltage loop produces the d-axis current reference, the q-axis reference is zero for unity power factor, and two inner PI current loops generate the converter voltage references with grid-voltage feed-forward and cross-coupling decoupling (ωLt). The loops are tuned in closed form:

- Current loop: crossover fc = min(Fsw/10, Fres/2.5), with the PI zero at fc/10 and the PWM delay Ta = 1/(2·Fsw) included: kpi = |1 + jTa·ωc| · |Rt + jωc·Lt| / |1 − jωi/ωc|, kii = ωi · kpi.
- DC-link voltage loop: crossover 150 Hz with the PI zero at 30 Hz: kpv = ωc2 · Cdc / (Kdc · |1 − jωi2/ωc2|), kiv = ωi2 · kpv, with Kdc = 1.5 · V(L-L) / Vdc.
- Modulation scaling: Kmod = 1000 / (Vdc/2) maps the controller output (volts) to the ±1000 PWM carrier; the modulation index is m = √(2/3) · V(L-L) / (Vdc/2) and must stay below 0.95.

2.7 Synchronous buck-boost converter

The maximum output voltage is 1.25 times the nominal battery voltage (500 V) and the maximum current is the power divided by the battery voltage:

$$Io,max = P / Vbat \qquad Dmax = Vo,max / (Vo,max + Vdc) \qquad IL = Io,max / (1 - Dmax) \qquad \Delta IL = 0.1 \cdot IL$$

$$L \geq Vdc \cdot Dmax / (\Delta IL \cdot fsw) \qquad Co \geq Io,max \cdot Dmax / (fsw \cdot \Delta Vo),\ \Delta Vo = 0.01 \cdot Vo,max$$

$$IL,peak = IL + \Delta IL/2 \qquad IL,SAT = 1.2 \cdot IL,peak \qquad Vsw,max = Vdc + Vo,max$$

L is rounded up to a multiple of 2 µH and Co to a multiple of 100 µF. The duty-cycle limit is Dlim = min(0.95, 1.13 · Dmax). The current-loop PI gains of the DC/DC stage are scaled from the proven base design with the inductor and the input voltage, kpb = 0.008 · (L/36 µH) · 1100/(Vdc + Vbat) and kib = 3.02 · (L/36 µH) · 1100/(Vdc + Vbat). The CV target is Vcv = 1.1175 · Vbat = 447 V.

2.8 Power, efficiency and power-factor calculation

The script logs the grid phase voltages and currents, the DC-link voltage and current, and the buck-boost output voltage and current, and calculates:

$$Pin = vaia + vbib + vcic \qquad Pdc = VdcIdc \qquad Pout = VbuckIbuck$$

$$\eta(AFE + LCL) = Pdc / Pin \qquad \eta(buck\text{-}boost) = Pout / Pdc \qquad \eta(overall) = Pout / Pin$$

Each power is averaged over one grid period (20 ms). The efficiencies are the means over the window 0.25–0.38 s, when the charging current is at its maximum in the CC stage; the energy efficiency is the integral of Pout divided by the integral of Pin from 0.05 s to the end of the run. The power factor is calculated over six whole grid periods (0.25–0.37 s): the real power is P = Σ(v·i), the apparent power is S = Σ(Vrms·Irms) over the three phases, the true power factor is P / S, the displacement power factor is the cosine of the angle between the 50 Hz voltage and current phasors, and the current distortion (THD) is the rms of the non-fundamental current divided by the fundamental current. The calculation counts only the modelled losses (switch and diode drops, filter and inductor resistance, damping branch) and excludes the internal loss of the battery.

2.9 Component values

Table 2 lists the main values calculated for the two designs used in this work. The complete lists, with formulas, are in Table 4, Table 8 and Appendix B.

Table 2: Component values and controller gains for the two designs (400 V / 200 kW and 220 V / 60 kW).

Quantity	400 V / 200 kW	220 V / 60 kW
DC-link voltage Vdc	700 V	400 V
DC-link capacitor Cdc	1031 µF	984 µF
Converter-side inductor L	107.2 µH	112.3 µH
Grid-side inductor Lf	27.1 µH	27.3 µH
Filter capacitor Cf	199 µF	197 µF
Resonant frequency Fres	2.43 kHz	2.42 kHz
Current loop: fc, kpi, kii	971 Hz, 0.852, 519.8	968 Hz, 0.882, 536.4
Voltage loop: fc2, kpv, kiv	150 Hz, 1.112, 209.5	150 Hz, 1.102, 207.8
Modulation scaling Kmod / index m	2.857 / 0.933	5.000 / 0.898
Maximum charging current Io,max	500 A	150 A
Maximum duty cycle Dmax / limit Dlim	0.417 / 0.471	0.556 / 0.628
Buck-boost inductor Lch (minimum)	36 µH (34.0 µH)	66 µH (65.8 µH)
Output capacitor Co (minimum)	500 µF (416.7 µF)	200 µF (166.7 µF)
Peak / saturation inductor current	900 A / 1080 A	354 A / 425 A
Switch stress Vsw,max / Isw,max	1200 V / 900 A	900 V / 354 A
DC/DC current loop kpb, kib	0.008, 3.02	0.0202, 7.61
CV target Vcv	447 V	447 V

The design checks pass for both designs (Fres between 6·Fs and 0.5·Fsw, and m below 0.95; m = 0.90 at 220 V is close to the limit). For 400 V / 200 kW the script warns that the 1200 V / 900 A switch stress exceeds a single 1200 V / 600 A module at the usual derating, so paralleled or higher-rated devices would be needed in hardware.

2.10 Assumptions

- The grid is a balanced, stiff three-phase 50 Hz sinusoidal source.
- The IGBTs and diodes are the standard Simscape Electrical models with their default on-state resistance and forward voltage, with no dead time and no thermal model; the buck-boost IGBT model includes fall and tail times that cause switching loss.
- The battery is the generic Lithium-Ion Battery block (400 V nominal, 250 Ah, 100 kWh), without temperature or aging effects; the maximum charging current is Io,max = P / Vbat (at most 500 A, i.e. 2C), and the power input is limited to 200 kW.
- Simulation settings: variable-step ode45 solver with a relative tolerance of 1e-3 and a discrete powergui sample time of 3 µs.
- The DC link starts precharged at its target voltage, and the PLL is assumed to track the grid after the first few milliseconds.
- The LCL damping branch keeps a 2 Ω resistor, and the filter and inductor resistances are 37 times the inductance (R = 37 L).
- The charging profile is a CC staircase of 40, 60, 80 and 100 % of Io,max in 0.07 s steps, with a time-based switch to a CV target of 447 V at 0.4 s, and a current-reference rate limit of 5 kA/s scaled with Io,max.

3. Results and Analysis

The results are given for the three cases requested (Cases 1–3, at 400 V and 200 kW) and for an additional case with a different grid voltage and power (Case 4, at 220 V and 60 kW). For every case the script prints all calculated parameters in the Command Window; they are presented as tables with the plots, and the complete printouts are in Appendix B. The comparison of the cases follows in Section 3.6, then the effect of the reference ramp, detailed waveforms, and the discussion of limitations.

3.1 Test cases

Table 3: The four simulated cases.

Case	Simulation time	Initial SOC	Grid and power
1	5 s	80 %	400 V / 200 kW
2	10 s	70 %	400 V / 200 kW
3	3 s	90 %	400 V / 200 kW
4 (additional)	0.5 s	70 %	220 V / 60 kW

3.2 Case 1: 5 s, initial SOC 80 %

Case 1 uses a simulation time of 5 s and an initial SOC of 80 % with the 400 V / 200 kW design. Table 4 lists the parameters calculated by the script and printed in the Command Window. They are also the design values of Cases 2 and 3, since the grid voltage and power are the same; only SOC0 and the simulation time differ. Figure 7 shows the complete scope and Figure 8 the power flow and efficiency.

Table 4: Parameters calculated by the script for 400 V, 200 kW and an initial SOC of 80 % (Command Window output).

Parameter	Value	Unit	Formula / meaning
INPUTS AND ASSUMPTIONS			
V_grid	400	V	grid line-to-line RMS (input)
P	2e+05	W	required charging power (input)
Vbat	400	V	battery nominal voltage (Battery block)
SOC0	80	%	initial battery state of charge (input, Battery block)
Fs	50	Hz	grid frequency
Fsw	10000	Hz	AFE switching frequency
fsw	1e+05	Hz	buck-boost switching frequency
I_line_rms	288.68	A	P / (sqrt(3) V_grid)
AFE RECTIFIER AND LCL FILTER			

Parameter	Value	Unit	Formula / meaning
Vdc_min	565.69	V	sqrt(2) V_grid
Vdc	700	V	ceil(1.2 Vdc_min / 50) * 50
Cdc	1031	uF	P / (240 dV Vph Fs), dV = 10 % of Vdc
I_ripple	81.65	A	0.2 sqrt(2) P / (sqrt(3) V_grid)
Zb	0.8	Ohm	V_grid^2 / P
Cb	3.9789	mF	1 / (Ws Zb)
L	107.17	uH	Vdc / (8 Fsw I_ripple)
Cf	198.94	uF	x Cb, x = 0.05
a	1683.3		L Cb Wsw^2
r	0.2525		from attenuation factor 5 %
Lf	27.06	uH	r L
Lt	134.22	uH	L + Lf
Wres	15253	rad/s	sqrt(Lt / (L Lf Cf))
Fres	2.4276	kHz	Wres / (2 pi)
rd_design	0.10985	Ohm	1 / (3 Wres Cf) (model keeps 2 Ohm)
R	0.0039651	Ohm	37 L
Rf	0.0010012	Ohm	37 Lf
Rt	0.0049663	Ohm	R + Rf
AFE CONTROLLER			
Ta	50	us	1 / (2 Fsw)
fc (current loop)	971.06	Hz	min(Fsw/10, Fres/2.5)
wi (PI zero)	610.13	rad/s	wc / 10
kpi	0.85198	V/A	|1+jTa wc| |Rt+j wc Lt| / |1-j wi/wc|
kii	519.82		wi kpi
fc2 (voltage loop)	150	Hz	chosen
wi2 (PI zero)	188.5	rad/s	wc2 / 5
Kdc	0.85714		1.5 V_grid / Vdc
kpv	1.1116		wc2 Cdc / (Kdc |1-j wi2/wc2|)
kiv	209.53		wi2 kpv
Kmod	2.8571		1000 / (Vdc/2), carrier is +/-1000
m_index	0.93314		sqrt(2/3) V_grid / (Vdc/2)

Parameter	Value	Unit	Formula / meaning
SYNCHRONOUS BUCK-BOOST			
Vo_max	500	V	1.25 Vbat
Io_max	500	A	P / Vbat
D_max	0.41667		Vo_max / (Vo_max + Vdc)
I_L	857.14	A	Io_max / (1 - D_max)
dIL	85.714	A	10 % of I_L
Lch_min	34.028	uH	Vdc D_max / (dIL fsw)
Lch	36	uH	rounded up to 2 uH
Co_min	416.67	uF	Io_max D_max / (fsw dVo), dVo = 1 %
Co	500	uF	rounded up to 100 uF
IL_peak	900	A	I_L + dIL/2
IL_SAT	1080	A	1.2 IL_peak
Vsw_max	1200	V	Vdc + Vo_max
Dlim	0.47083		min(0.95, 1.13 D_max)
kpb	0.008		0.008 (Lch/36u) (1100/(Vdc+Vbat))
kib	3.0202		3.02 (Lch/36u) (1100/(Vdc+Vbat))
Vcv	447	V	1.1175 Vbat (constant-voltage target)

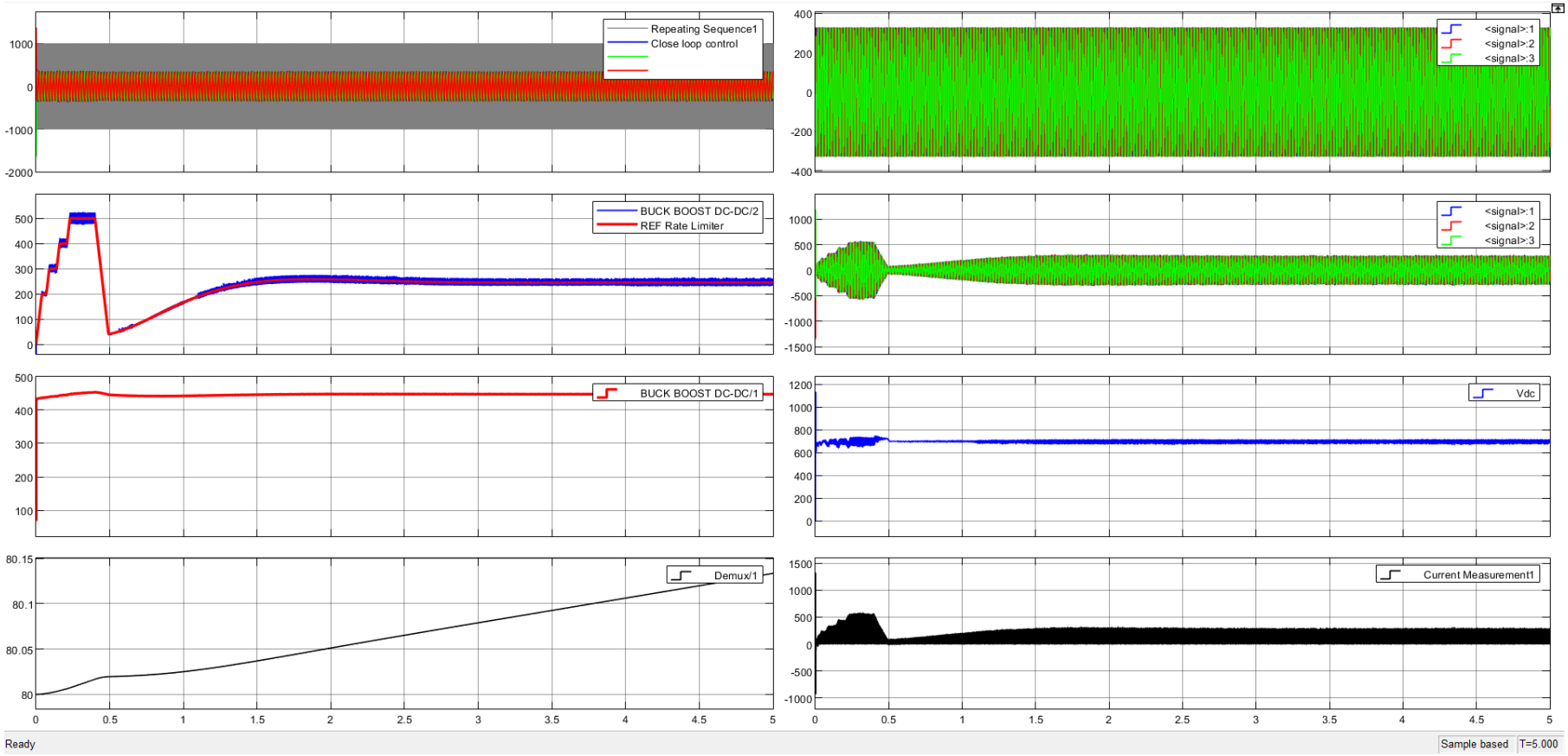


Figure 7: Complete scope for Case 1 (5 s, initial SOC 80 %).

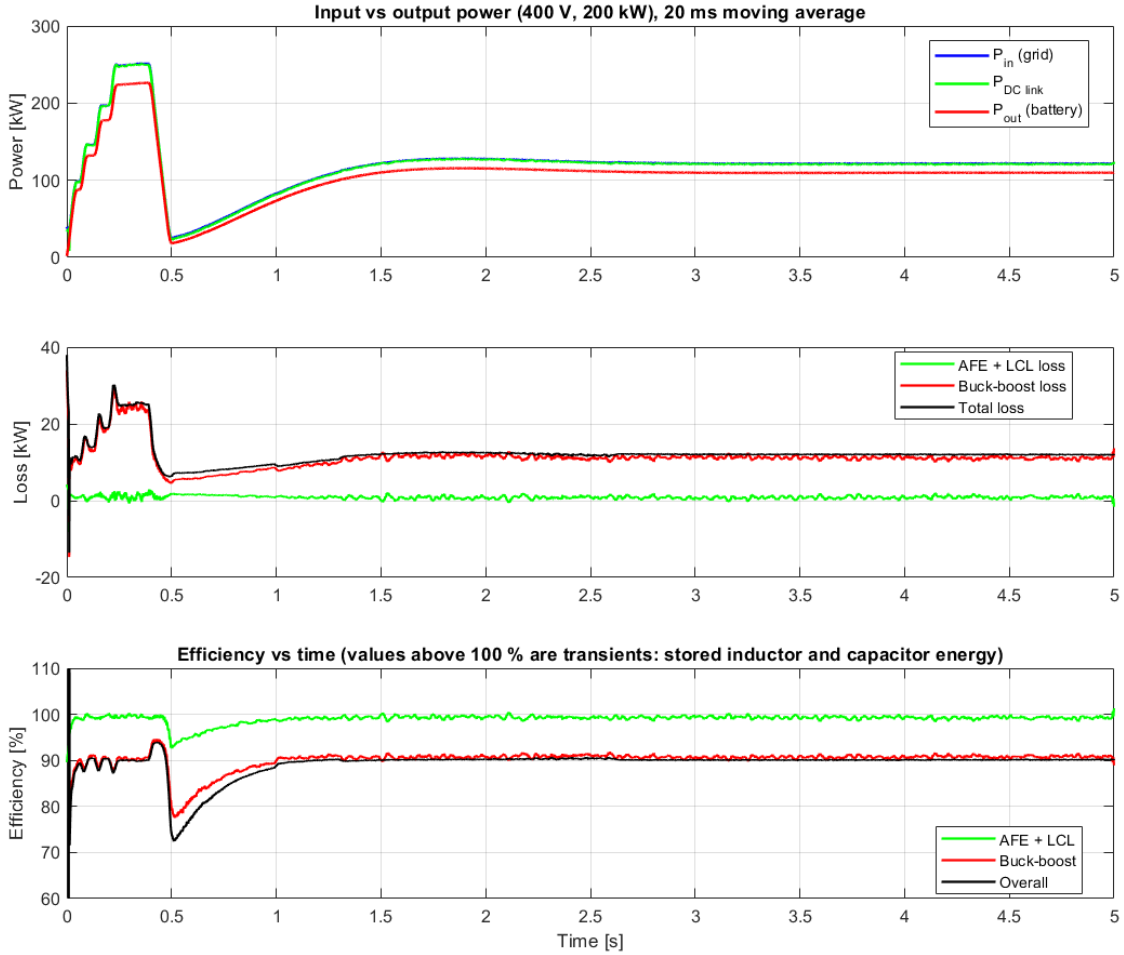


Figure 8: Input and output power, losses and efficiency for Case 1.

Table 5: Key results for Case 1 (powers and efficiencies averaged over 0.25–0.38 s, power factor over 0.25–0.37 s).

Quantity	**Value**
SOC at start / end / change	80 % / 80.134 % / +0.134 %
Grid input power	250.2 kW

Quantity	Value
DC-link power	249.2 kW
Battery (output) power	225.1 kW
AFE + LCL efficiency	99.6 %
Buck-boost efficiency	90.3 %
Overall efficiency	90.0 %
Energy efficiency (0.05–5 s)	89.9 %
True power factor / displacement power factor	0.999 / 1.000
Voltage–current angle	−0.02°
Fundamental grid current	360.9 A rms
Current distortion (THD)	4.3 %

The main observations for Case 1 are:

- The SOC rises steadily from 80 % to 80.134 % over the 5 s run, with a steeper slope in the CC stage (up to 0.4 s) and a steady slope in the CV stage.
- The charging current follows the staircase reference up to about 500 A, holds there until 0.4 s, falls to about 40 A when the converter changes to CV mode, and then recovers smoothly to a steady value of about 250 A that is held until 5 s.
- The charging voltage rises to about 452 V at 0.4 s and then settles near the 447 V CV target for the rest of the run, so the CV loop regulates the battery voltage.
- The DC link settles at about 700 V after the first 0.5 s, with a larger ripple while the current steps up and a small ripple afterwards.
- The grid power follows the battery power: about 250 kW at the CC peak and about 120 kW (about 110 kW into the battery) in the steady CV stage. The overall efficiency is about 90 % throughout, and the buck-boost stage causes almost all of the loss.
- In the full-current CC stage the battery receives 225 kW against the 200 kW requested (Section 3.9).

3.3 Case 2: 10 s, initial SOC 70 %

Case 2 uses a simulation time of 10 s and an initial SOC of 70 % with the same 400 V / 200 kW design, so the calculated parameters are those of Table 4 with SOC0 = 70 %. Figure 9 shows the complete scope and Figure 10 the power flow and efficiency.

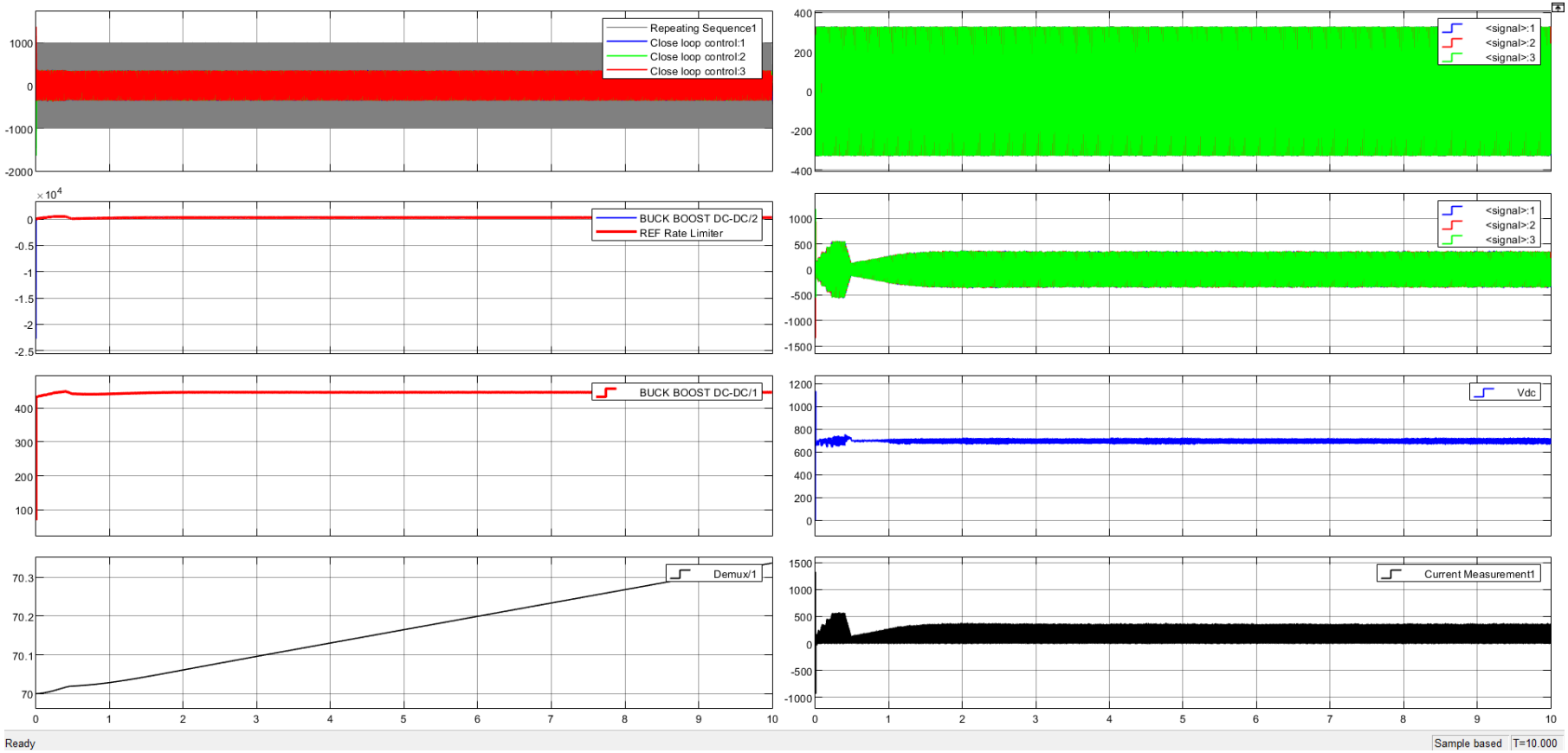


Figure 9: Complete scope for Case 2 (10 s, initial SOC 70 %).

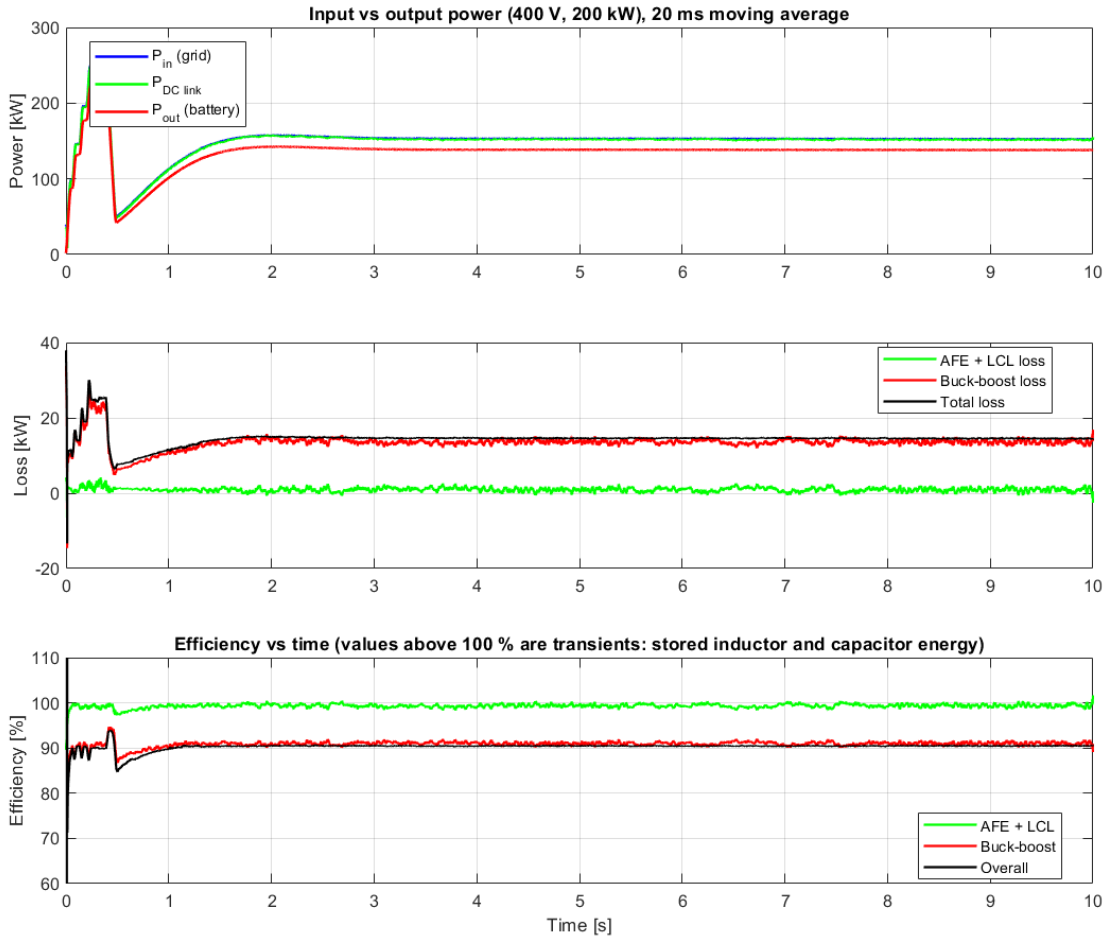


Figure 10: Input and output power, losses and efficiency for Case 2.

Table 6: Key results for Case 2 (powers and efficiencies averaged over 0.25–0.38 s, power factor over 0.25–0.37 s).

Quantity	**Value**
SOC at start / end / change	70 % / 70.337 % / +0.337 %
Grid input power	248.6 kW
DC-link power	246.5 kW

Quantity	Value
Battery (output) power	223.6 kW
AFE + LCL efficiency	99.2 %
Buck-boost efficiency	90.7 %
Overall efficiency	90.0 %
Energy efficiency (0.05–10 s)	90.4 %
True power factor / displacement power factor	0.999 / 1.000
Voltage–current angle	−0.02°
Fundamental grid current	358.6 A rms
Current distortion (THD)	4.3 %

The main observations for Case 2 are:

- The SOC rises from 70 % to 70.337 % over the 10 s run, with a steeper slope in the CC stage and a steady slope of about 0.032 % per second in the CV stage.
- The CC stage is the same as in the other 400 V cases: the current reaches about 500 A at 0.23 s and holds until 0.4 s, and the battery receives about 224 kW against the 200 kW requested.
- After the switch to CV mode the charging voltage settles near 447 V for the remaining 9.5 s. The grid power settles at about 152 kW (about 138 kW into the battery) from about 3 s, which corresponds to a charging current of about 310 A; the SOC slope (about 290 A) agrees. These are estimates from power and SOC rate, not direct readings of the battery current.
- The CV current is higher than in Case 1 (about 250 A at 80 % SOC) because the battery at 70 % SOC is further below the 447 V target, so the CV loop demands more current.
- The DC link stays at about 700 V for the whole run with a small ripple and no drift, and the efficiency is steady at about 90.5 % overall in the CV stage (about 99.5 % for the AFE and filter and about 91 % for the buck-boost stage).

3.4 Case 3: 3 s, initial SOC 90 %

Case 3 uses a simulation time of 3 s and an initial SOC of 90 % with the same 400 V / 200 kW design, so the calculated parameters are those of Table 4 with SOC0 = 90 %. Figure 11 shows the complete scope and Figure 12 the power flow and efficiency.

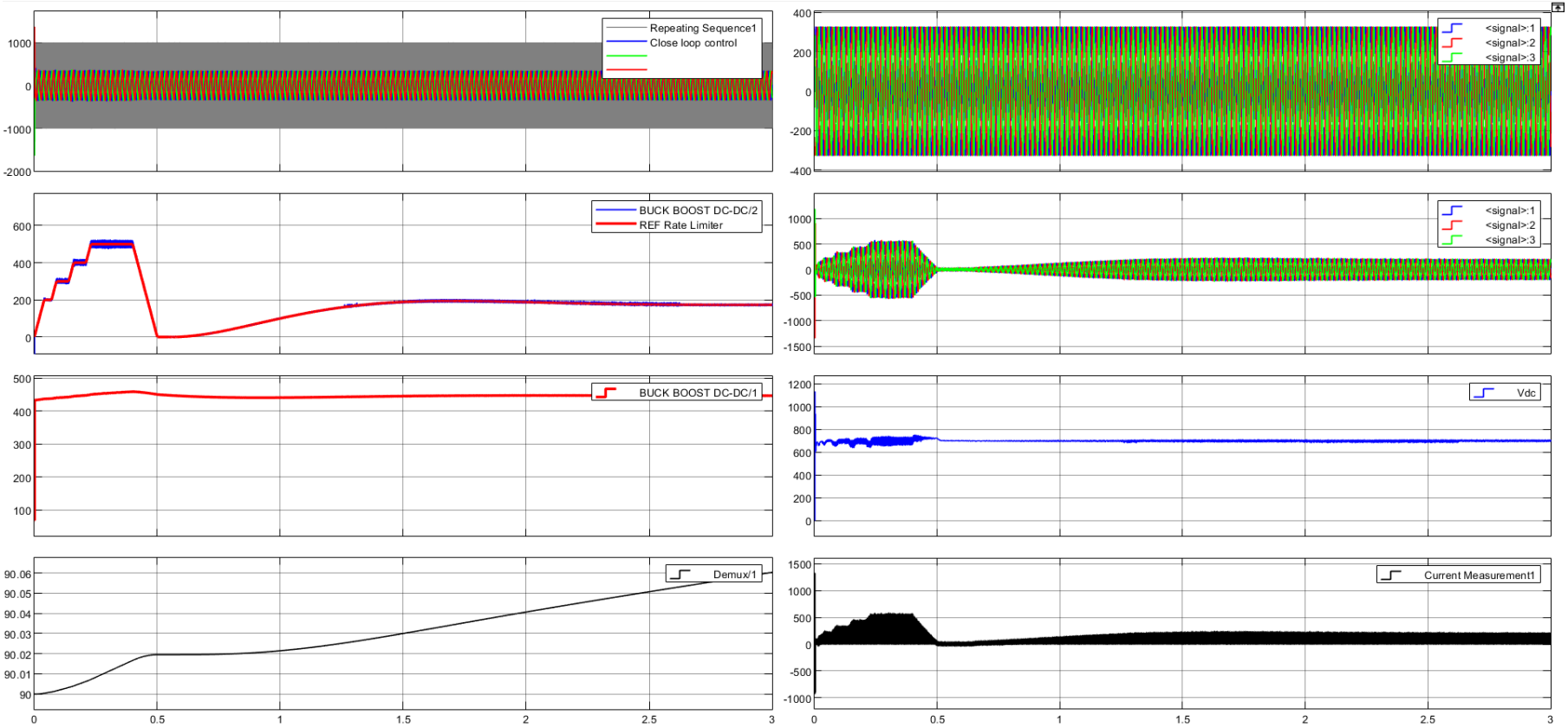


Figure 11: Complete scope for Case 3 (3 s, initial SOC 90 %).

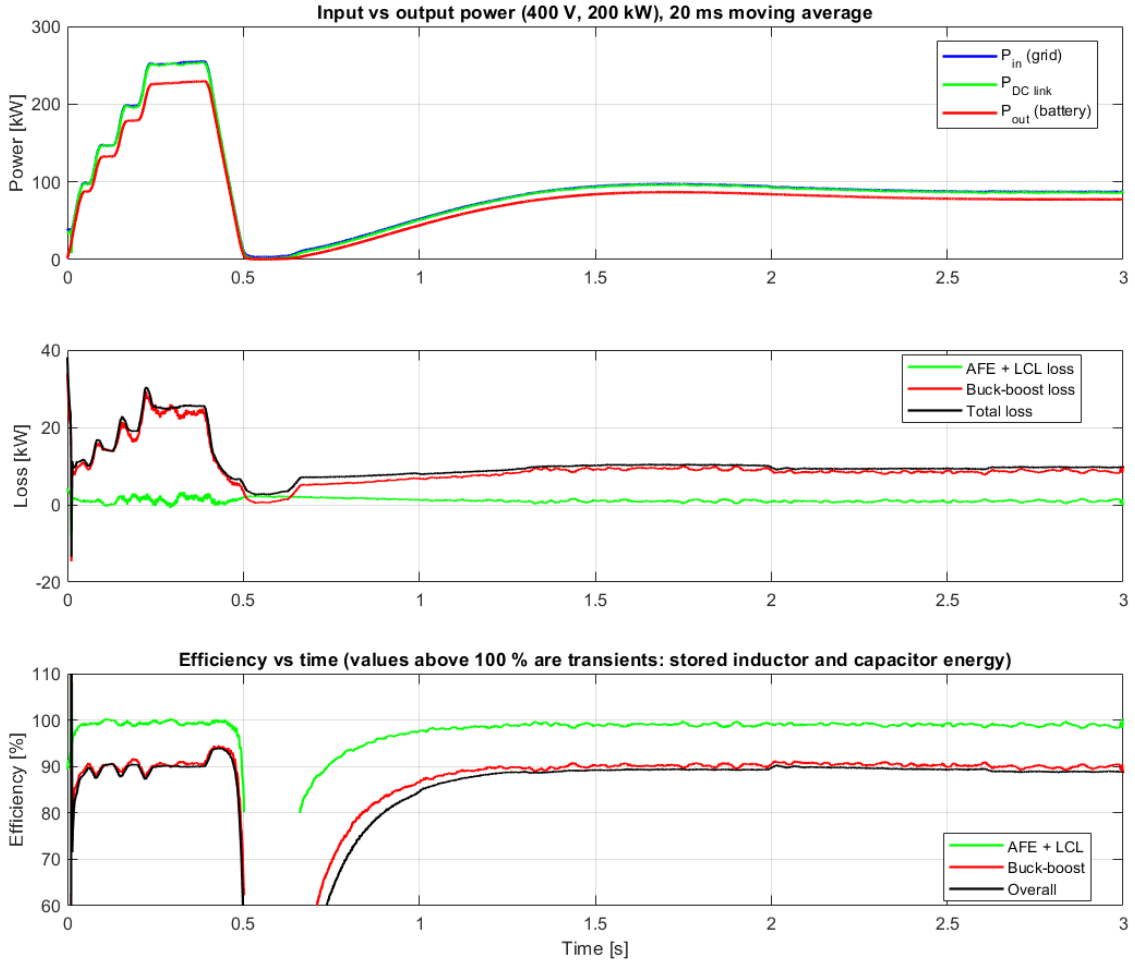


Figure 12: Input and output power, losses and efficiency for Case 3.

Table 7: Key results for Case 3 (powers and efficiencies averaged over 0.25–0.38 s, power factor over 0.25–0.37 s).

Quantity	**Value**
SOC at start / end / change	90 % / 90.060 % / +0.060 %
Grid input power	252.9 kW

Quantity	Value
DC-link power	251.5 kW
Battery (output) power	227.6 kW
AFE + LCL efficiency	99.5 %
Buck-boost efficiency	90.5 %
Overall efficiency	90.0 %
Energy efficiency (0.05–3 s)	88.7 %
True power factor / displacement power factor	0.999 / 1.000
Voltage–current angle	−0.02°
Fundamental grid current	364.8 A rms
Current distortion (THD)	4.2 %

The main observations for Case 3 are:

- The SOC rises from 90 % to 90.060 % in 3 s. It rises during the CC stage, stays flat for about 0.3 s after the switch to CV mode, and then rises again at a lower rate.
- The CC stage still reaches about 500 A, because the CC reference does not depend on the battery voltage. Because the battery is almost full, its voltage rises to about 460 V by 0.4 s, above the 447 V CV target, and the battery receives about 228 kW.
- When the converter changes to CV mode at 0.4 s, the CV loop sees a voltage above its target and reduces the charging current to almost zero for roughly 0.15 s (0.5 to 0.65 s). As the battery voltage relaxes towards 447 V the current recovers slowly and settles at about 175 A, with about 88 kW from the grid and 78 kW into the battery at 3 s.
- The efficiency is about 90 % in the CC stage. In the low-power interval after 0.5 s the efficiency plot has a gap, because the power is below 5 % of the rated value, and the efficiency dips while the power recovers; by 1.5 s it settles at about 89 % overall. This is why the energy efficiency over the run (88.7 %) is lower than in the other cases.
- The DC link settles at about 700 V once the large current steps have ended.

3.5 Case 4 (additional): 0.5 s, initial SOC 70 %, 220 V / 60 kW

Case 4 shows that the design adapts to a different grid voltage and power: the grid is 220 V line-to-line, the required power is 60 kW, the initial SOC is 70 % and the simulation time is 0.5 s. The script calculates a 400 V DC link, a smaller current limit of 150 A and new filter, inductor and capacitor values (Table 8). Figure 13 shows the complete scope and Figure 14 the power flow and efficiency.

Table 8: Parameters calculated by the script for 220 V, 60 kW and an initial SOC of 70 % (Command Window output).

Parameter	Value	Unit	Formula / meaning
INPUTS AND ASSUMPTIONS			
V_grid	220	V	grid line-to-line RMS (input)
P	60000	W	required charging power (input)
Vbat	400	V	battery nominal voltage (Battery block)
SOC0	70	%	initial battery state of charge (input, Battery block)
Fs	50	Hz	grid frequency
Fsw	10000	Hz	AFE switching frequency
fsw	1e+05	Hz	buck-boost switching frequency
I_line_rms	157.46	A	P / (sqrt(3) V_grid)
AFE RECTIFIER AND LCL FILTER			
Vdc_min	311.13	V	sqrt(2) V_grid
Vdc	400	V	ceil(1.2 Vdc_min / 50) * 50
Cdc	984.12	uF	P / (240 dV Vph Fs), dV = 10 % of Vdc
I_ripple	44.536	A	0.2 sqrt(2) P / (sqrt(3) V_grid)
Zb	0.80667	Ohm	V_grid^2 / P
Cb	3.946	mF	1 / (Ws Zb)
L	112.27	uH	Vdc / (8 Fsw I_ripple)
Cf	197.3	uF	x Cb, x = 0.05
a	1748.9		L Cb Wsw^2
r	0.24292		from attenuation factor 5 %
Lf	27.273	uH	r L
Lt	139.54	uH	L + Lf
Wres	15198	rad/s	sqrt(Lt / (L Lf Cf))
Fres	2.4189	kHz	Wres / (2 pi)
rd_design	0.11116	Ohm	1 / (3 Wres Cf) (model keeps 2 Ohm)
R	0.0041539	Ohm	37 L
Rf	0.0010091	Ohm	37 Lf
Rt	0.005163	Ohm	R + Rf
AFE CONTROLLER			
Ta	50	us	1 / (2 Fsw)
fc (current loop)	967.55	Hz	min(Fsw/10, Fres/2.5)

Parameter	Value	Unit	Formula / meaning
wi (PI zero)	607.93	rad/s	wc / 10
kpi	0.88226	V/A	|1+jTa wc| |Rt+j wc Lt| / |1-j wi/wc|
kii	536.35		wi kpi
fc2 (voltage loop)	150	Hz	chosen
wi2 (PI zero)	188.5	rad/s	wc2 / 5
Kdc	0.825		1.5 V_grid / Vdc
kpv	1.1024		wc2 Cdc / (Kdc |1-j wi2/wc2|)
kiv	207.8		wi2 kpv
Kmod	5		1000 / (Vdc/2), carrier is +/-1000
m_index	0.89815		sqrt(2/3) V_grid / (Vdc/2)
SYNCHRONOUS BUCK-BOOST			
Vo_max	500	V	1.25 Vbat
Io_max	150	A	P / Vbat
D_max	0.55556		Vo_max / (Vo_max + Vdc)
I_L	337.5	A	Io_max / (1 - D_max)
dIL	33.75	A	10 % of I_L
Lch_min	65.844	uH	Vdc D_max / (dIL fsw)
Lch	66	uH	rounded up to 2 uH
Co_min	166.67	uF	Io_max D_max / (fsw dVo), dVo = 1 %
Co	200	uF	rounded up to 100 uF
IL_peak	354.38	A	I_L + dIL/2
IL_SAT	425.25	A	1.2 IL_peak
Vsw_max	900	V	Vdc + Vo_max
Dlim	0.62778		min(0.95, 1.13 D_max)
kpb	0.020167		0.008 (Lch/36u) (1100/(Vdc+Vbat))
kib	7.6135		3.02 (Lch/36u) (1100/(Vdc+Vbat))
Vcv	447	V	1.1175 Vbat (constant-voltage target)

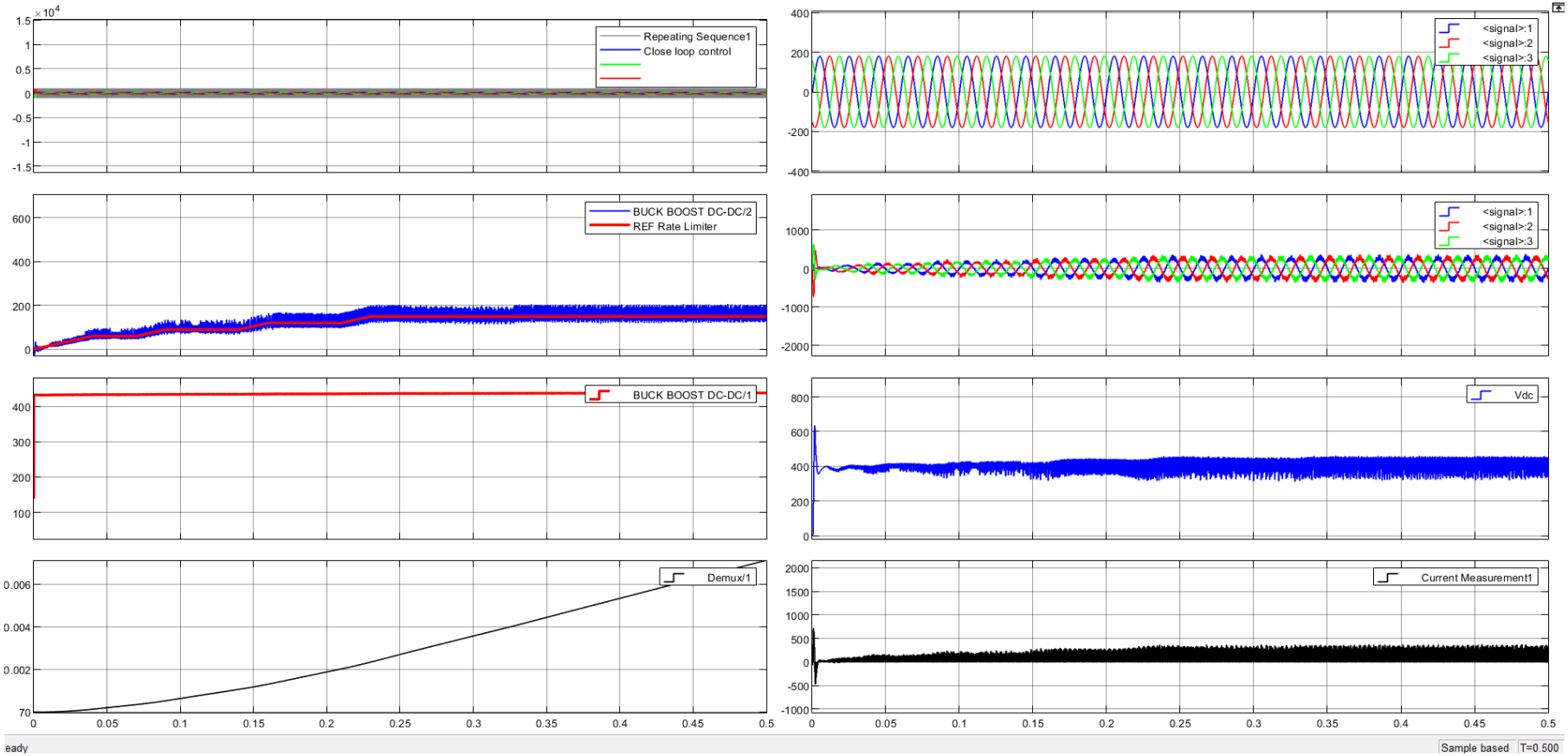


Figure 13: Complete scope for Case 4 (0.5 s, initial SOC 70 %, 220 V, 60 kW).

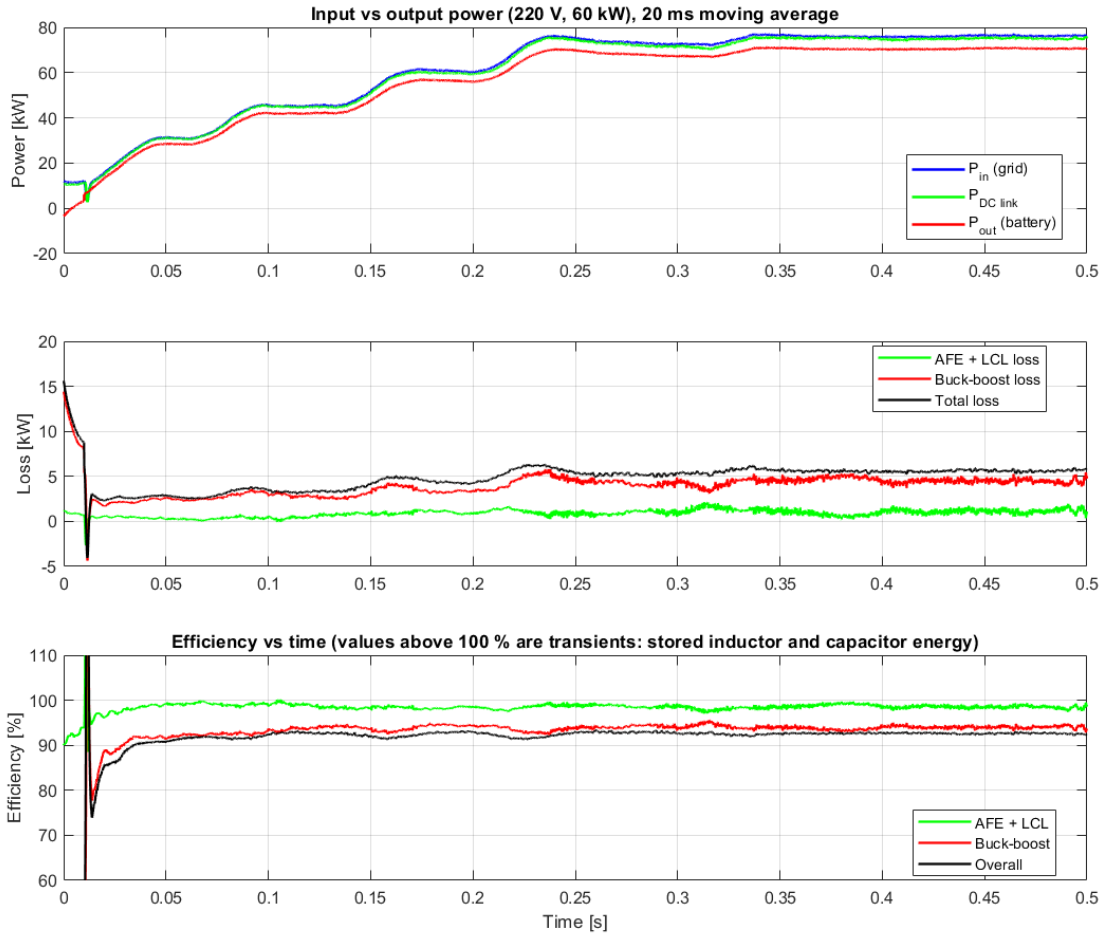


Figure 14: Input and output power, losses and efficiency for Case 4.

Table 9: Key results for Case 4 (powers and efficiencies averaged over 0.25–0.38 s, power factor over 0.25–0.37 s).

Quantity	**Value**
SOC at start / end / change	70 % / 70.007 % / +0.007 %
Grid input power	74.4 kW
DC-link power	73.4 kW

Quantity	Value
Battery (output) power	69.0 kW
AFE + LCL efficiency	98.5 %
Buck-boost efficiency	94.0 %
Overall efficiency	92.7 %
Energy efficiency (0.05–0.5 s)	92.5 %
True power factor / displacement power factor	0.988 / 1.000
Voltage–current angle	−0.03°
Fundamental grid current	195.0 A rms
Current distortion (THD)	15.5 %

The main observations for Case 4 are:

- The grid voltage peak is about 180 V (the phase peak of 220 V line-to-line), and the DC link is regulated around the 400 V reference.
- The SOC rises from 70 % to 70.007 %, which agrees with a charging current that follows the staircase up to 150 A (about 0.007 % in 0.5 s for a 250 Ah battery).
- The charging current follows the reference up to about 150 A, with a wider switching ripple than at 400 V. The grid current does not fall at 0.4 s, which is consistent with the CV loop output being limited at the 150 A maximum; this was not confirmed from the battery-current signal.
- The efficiencies are higher than at 400 V (92.7 % overall, 98.5 % for the AFE and filter and 94.0 % for the buck-boost stage), and the buck-boost loss is much smaller (4.4 kW), consistent with the lower inductor current (about 340 A instead of about 860 A).
- The battery receives about 69 kW and the grid supplies about 74 kW, above the 60 kW requested (Section 3.9).
- The performance is not as good as at 400 V. The DC-link voltage shows a ripple of roughly 100 V peak-to-peak by the end of the run (about 25 % of 400 V, against the 10 % design target), and the current distortion is 15.5 % against 4.3 % at 400 V, which lowers the true power factor to 0.988 even though the displacement power factor is 1.000.

3.6 Comparison of the four cases

Table 10: Comparison of the four cases. The CV-stage power and current are read from the plots (battery power divided by 447 V); Case 4 ends before a clear CV stage.

Quantity	Case 1	Case 2	Case 3	Case 4
Grid and power	400 V / 200 kW	400 V / 200 kW	400 V / 200 kW	220 V / 60 kW
Simulation time	5 s	10 s	3 s	0.5 s

Quantity	Case 1	Case 2	Case 3	Case 4
Initial SOC	80 %	70 %	90 %	70 %
SOC at end	80.134 %	70.337 %	90.060 %	70.007 %
SOC change	+0.134 %	+0.337 %	+0.060 %	+0.007 %
Grid input power (0.25–0.38 s)	250.2 kW	248.6 kW	252.9 kW	74.4 kW
Battery power (0.25–0.38 s)	225.1 kW	223.6 kW	227.6 kW	69.0 kW
Battery power vs request	+12.5 %	+11.8 %	+13.8 %	+15.0 %
AFE + LCL efficiency	99.6 %	99.2 %	99.5 %	98.5 %
Buck-boost efficiency	90.3 %	90.7 %	90.5 %	94.0 %
Overall efficiency	90.0 %	90.0 %	90.0 %	92.7 %
Energy efficiency	89.9 %	90.4 %	88.7 %	92.5 %
True power factor	0.999	0.999	0.999	0.988
Displacement power factor	1.000	1.000	1.000	1.000
Current distortion (THD)	4.3 %	4.3 %	4.2 %	15.5 %
Battery power at end (CV)	about 110 kW	about 138 kW	about 78 kW	not reached
CV charging current (estimate)	about 250 A	about 310 A	about 175 A	not reached

The CC stage is nearly the same in the three 400 V cases: 224–228 kW into the battery, a displacement power factor of 1.000 and an overall efficiency of about 90 %. The cases differ in the CV stage. The higher the initial SOC, the closer the battery voltage is to the 447 V target and the lower the charging current: about 310 A at 70 % SOC, about 250 A at 80 % and about 175 A at 90 %. This is why the SOC increase per second falls as the battery fills, as expected for a CC/CV charger. Case 4 has the highest efficiency because its currents are lower, but it also has the lowest true power factor and the highest current distortion, which result from the lower DC-link voltage and the larger switching ripple.

3.7 Effect of the reference ramp

Figure 15 shows the first 0.5 s at 400 V and 200 kW without the ramp blocks and Figure 16 shows the same run with the ramp blocks (both runs started at 70 % SOC; the run without the ramp was made before SOC0 became a variable). Without the ramp, the current reference steps instantly and the DC-link voltage overshoots to about 1000 V when the charger switches from CC to CV at 0.4 s. With the ramp, the charging current and the grid current rise and fall smoothly and the DC link stays around 700–750 V at that instant. A short start-up transient at t = 0 remains in both cases.

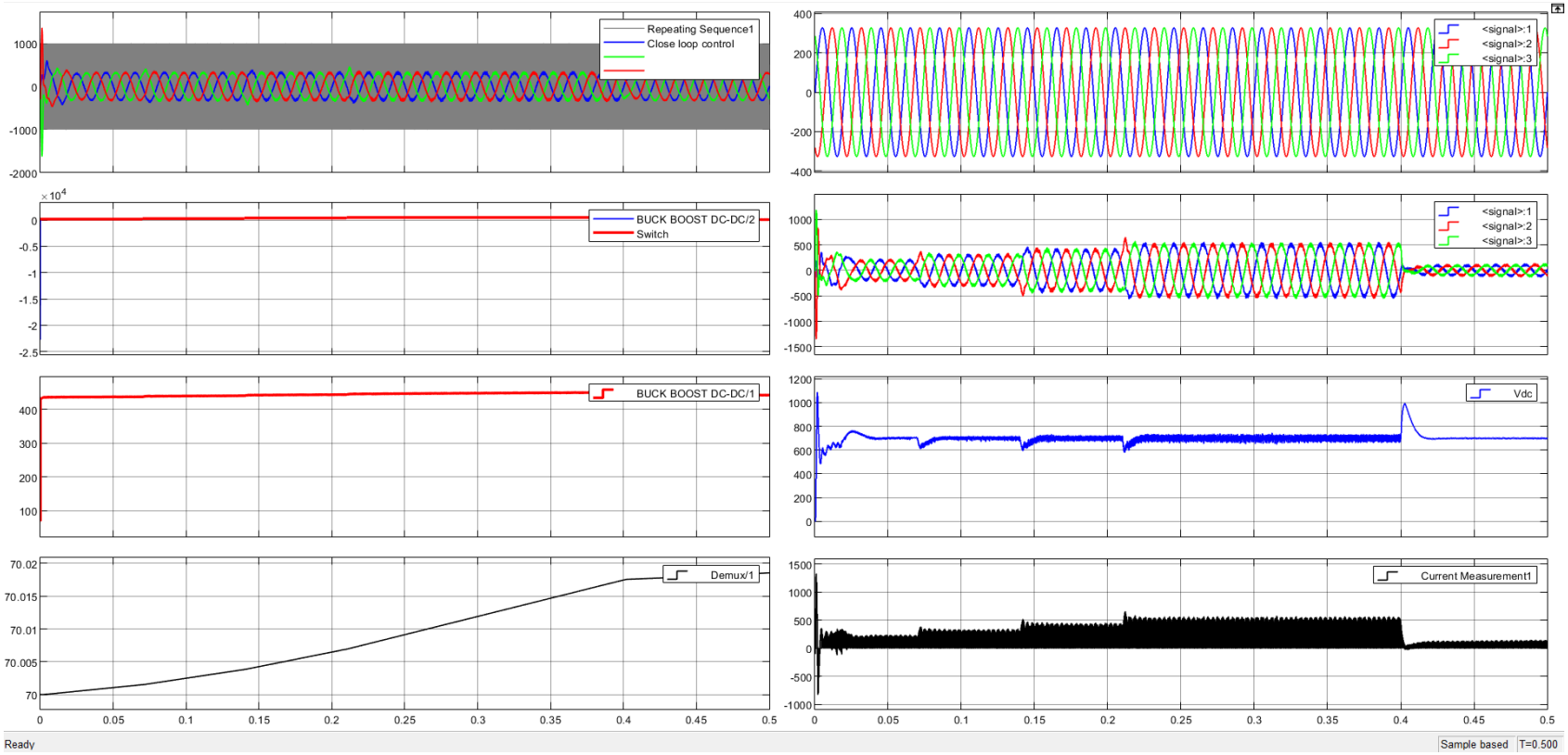


Figure 15: Results for the first 0.5 s without the reference ramp (400 V, 200 kW).

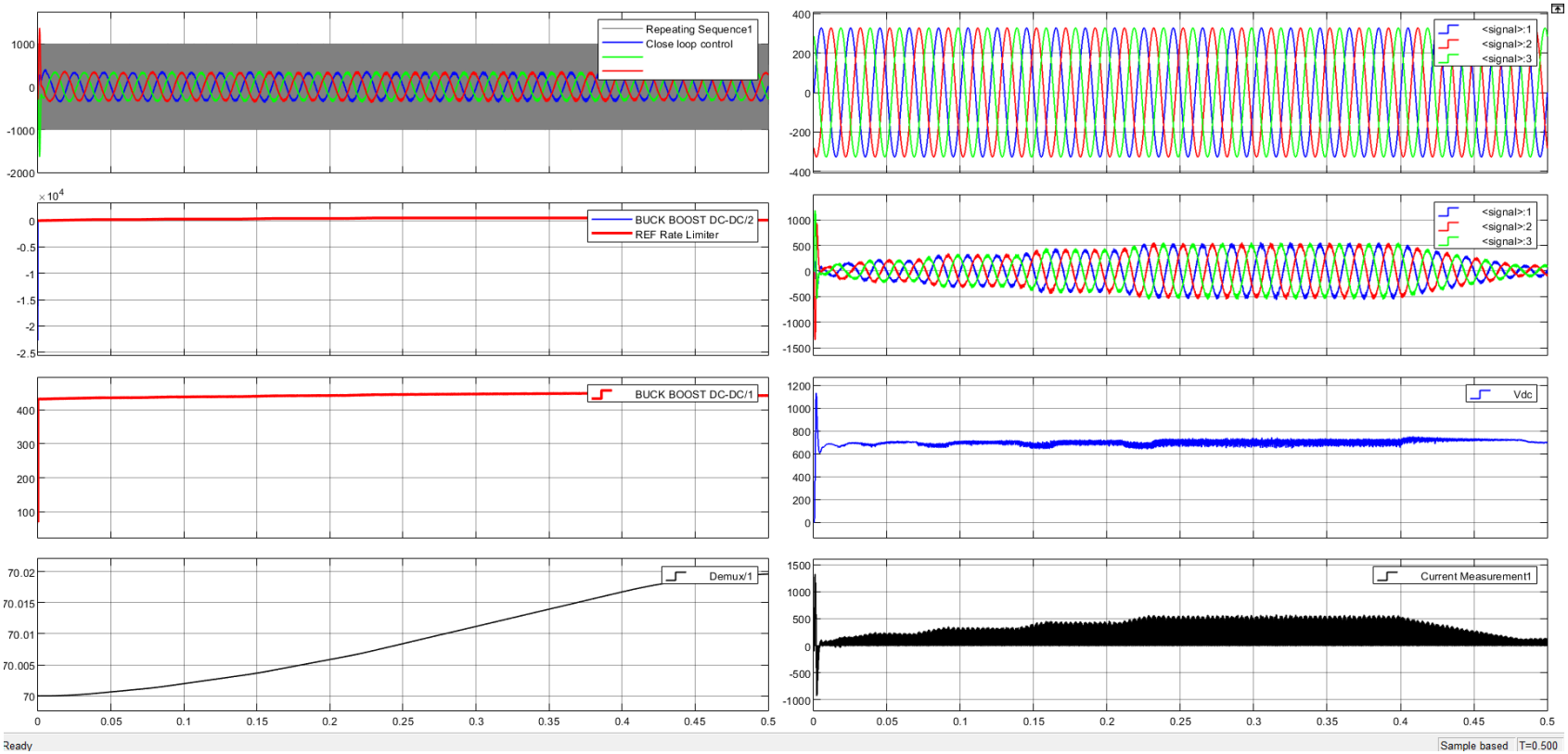


Figure 16: Results for the first 0.5 s with the reference ramp (400 V, 200 kW).

3.8 Detailed waveforms and unity power factor

Figure 17 to Figure 19 are zoomed views of a 0.5 s run at 400 V and 200 kW with 70 % SOC, equivalent to the first 0.5 s of Case 2. Figure 17 shows the charging current (reference after the rate limiter in red, measured current in blue): it follows the reference through the 200, 300, 400 and 500 A levels, holds at about 500 A until 0.4 s and then falls gradually as the converter changes to CV mode. Figure 18 shows the grid currents and the DC-link voltage. The start-up transient is visible at the left: the grid current reaches about 1300 A and the DC-link voltage about 1100 V, with a dip to about 600 V, and the system settles within a few tens of milliseconds. Figure 19 shows full-current operation between 0.25 s and 0.31 s: the grid currents are three-phase sinusoids with a peak of about 500 A and a visible switching ripple, the DC-link voltage stays around 700 V with a ripple of about ±35 V, and the DC-link current peaks at about 500 A. Comparing the top two panels, the voltage and current of each phase peak at the same instants (phase 1 at about 0.265 s, phase 2 at about 0.252 s and phase 3 at about 0.258 s), so the grid current is in phase with the grid voltage. The calculated displacement power factor of 1.000 (Table 10) confirms this.

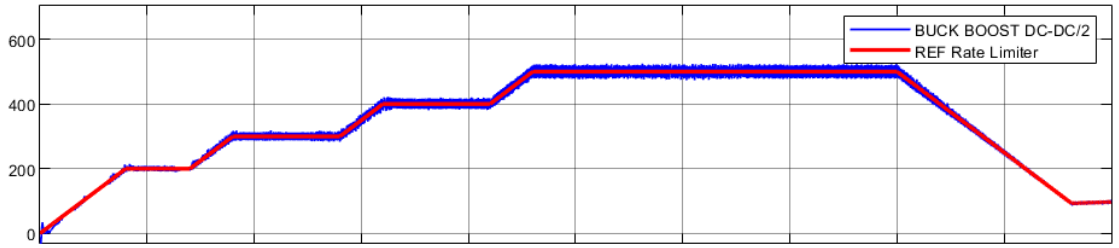


Figure 17: Charging current and its reference (400 V, 200 kW).

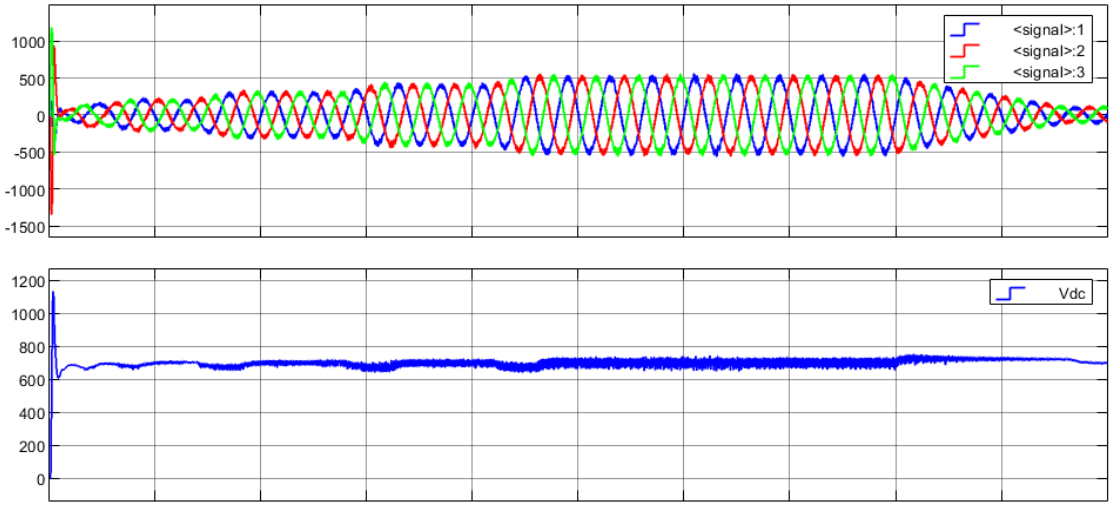


Figure 18: Grid currents and DC-link voltage, including the start-up transient (400 V, 200 kW).

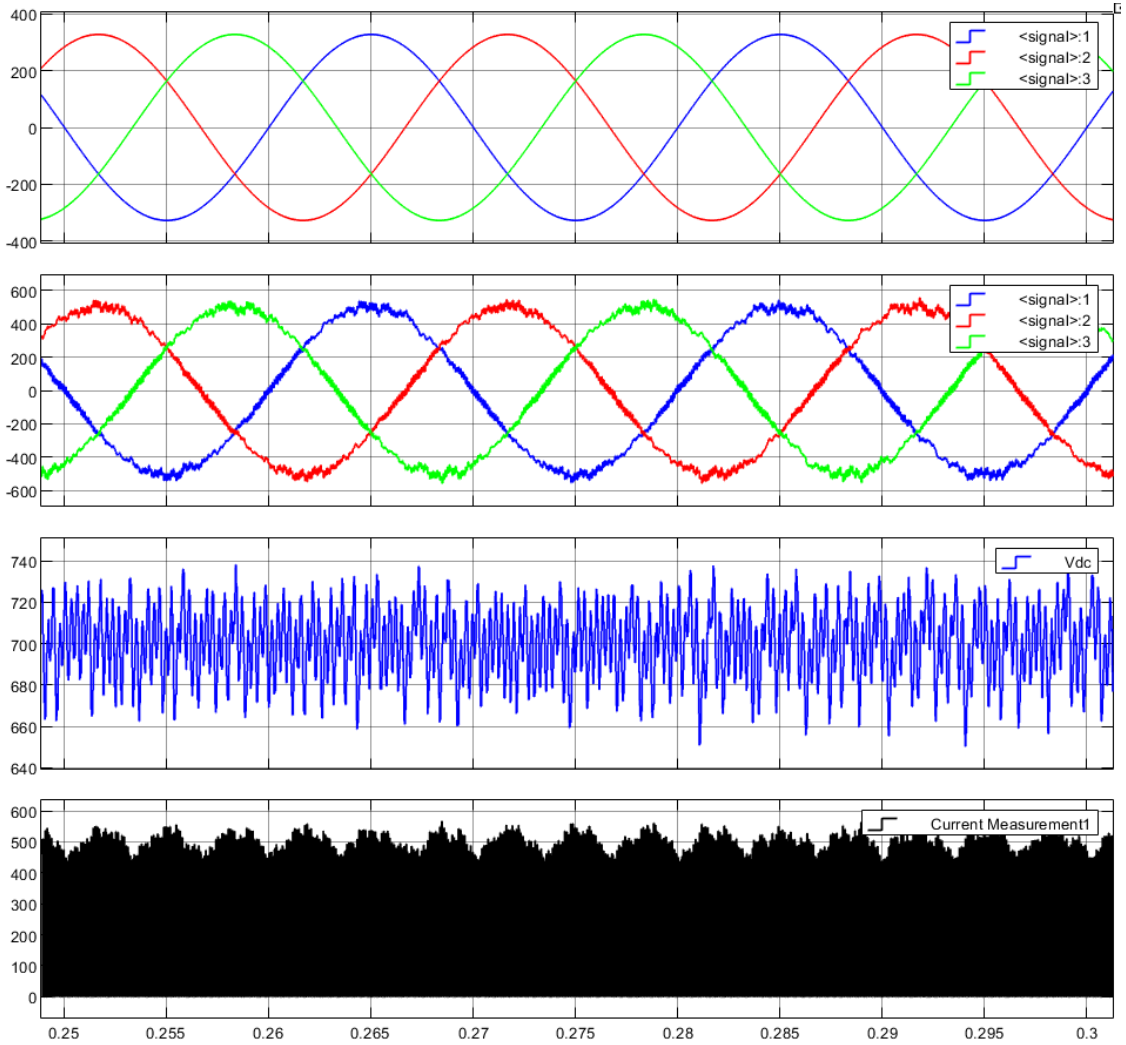


Figure 19: Grid voltages (top), grid currents, DC-link voltage and DC-link current at full current, 0.25–0.31 s (400 V, 200 kW).

3.9 Discussion and limitations

- Power above the request: in the full-current CC stage the battery receives 224–228 kW at 400 V / 200 kW (11.8–13.8 % above the request) and 69 kW at 220 V / 60 kW (15 % above), and the grid supplies about 249–253 kW and 74 kW. The maximum current is sized with the nominal battery voltage ($I_{o,max} = P / V_{bat}$, 500 A), but the battery terminal voltage is about 447–460 V at that current. Sizing the current limit with the CV voltage (P / V_{cv}, about 447 A at 200 kW) would give the requested power.
- Buck-boost efficiency: at 400 V the DC/DC stage is only about 90.5 % efficient and causes almost all of the loss (22.8–24.1 kW). This is probably dominated by the switching loss of the IGBT model, whose turn-off times (1 µs fall and 2 µs tail) are significant at 100 kHz and about 900 A; it was not isolated by a separate loss breakdown. A SiC MOSFET module would be more suitable at this frequency.
- CV switching: the change from CC to CV mode happens at a fixed time (0.4 s), not when the battery voltage reaches its limit. At 90 % SOC (Case 3) the battery voltage reached about 460 V by 0.4 s, above the 447 V target, so the CV loop cut the charging current to almost zero for roughly 0.15 s before it recovered. A voltage-triggered switch would avoid this.
- Start-up transient: at t = 0 the DC-link voltage and the grid current show a spike lasting a few milliseconds before the PLL and the controllers settle. It does not persist. Delaying the PWM enable until the PLL has locked would remove it.
- Low-voltage operation: at 220 V / 60 kW the DC link is only 400 V and the modulation index is 0.90, close to the linear limit. The DC-link ripple (about 25 % peak-to-peak) is larger than the 10 % design target, and the current distortion is 15.5 % against 4.3 % at 400 V. The gains are calculated for each operating point but were tuned and verified mainly at 400 V / 200 kW, so low-voltage operation needs further tuning.
- Switch stress: at 400 V / 200 kW the converter switches see 1200 V and 900 A, so a single 1200 V / 600 A module is insufficient for a real implementation. At higher grid voltages the DC link rises with the grid voltage and the stress grows further.
- Battery: the Battery block is a fixed 400 V / 250 Ah (100 kWh) pack, so the power input is limited to 200 kW (2C). Other battery voltages or capacities require changing the battery block. The absolute SOC change is small because the simulated times (0.5–10 s) are short compared with a real charge.
- Damping resistor: the shunt filter branch keeps a damping resistance of 2 Ω, while the design equation gives about 0.11 Ω. The model with 2 Ω was used for all results; the effect of the calculated value was not evaluated.
- Efficiency and power factor: the efficiency includes only the modelled losses; battery internal loss, magnetic core loss, gate-drive power, dead time and detailed switching-loss models are not included, so the values are probably higher than a hardware measurement. The true power factor is reduced by the current distortion (which includes switching ripple), not by a phase shift.
- Preliminary checking: before the Simulink runs, the AFE gains were checked with an independent switching-level simulation of the rectifier and LCL filter written outside Simulink, at several grid voltages and powers. The complete system was simulated in Simulink in the four cases of this report.

4. Conclusion

A dual-stage EV charger of up to 200 kW, with an AFE rectifier and LCL filter followed by a synchronous buck-boost converter, was designed and simulated in MATLAB/Simulink. A MATLAB script calculates all

component values and controller gains from the grid voltage, the required power and the initial battery SOC, prints each calculation in the Command Window, runs the model and reports the power, efficiency and power factor. In the three cases at 400 V and 200 kW (5 s at 80 % SOC, 10 s at 70 % and 3 s at 90 %) the DC link was regulated at 700 V, the battery was charged in CC mode and then CV mode at 447 V, and the SOC increased by 0.134 %, 0.337 % and 0.060 %. The grid-side displacement power factor was 1.000 with a true power factor of 0.999, and the overall efficiency was about 90 % (99.2–99.6 % for the AFE and filter and 90.3–90.7 % for the buck-boost stage). The additional case at 220 V and 60 kW (0.5 s) also increased the SOC, with an overall efficiency of 92.7 % and a true power factor of 0.988. The main limitations are that the battery receives 12–15 % more power than requested in the CC stage, that the CC-to-CV switch is time-based so the battery voltage exceeded the 447 V target before the switch at 90 % SOC, that the IGBT buck-boost stage limits the efficiency at 100 kHz, and that a short start-up transient and a larger ripple at 220 V remain.

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Appendix A: MATLAB Design Script (charger_design_RAMP_v3.m)

```
%% ======================================================================
%  DUAL-STAGE EV CHARGER  -  design + simulation + power/efficiency report
%  Inputs : grid line-to-line RMS voltage and required charging power.
%  Does   : 1) calculates every design variable and prints each one,
%              with its formula, in the Command Window;
%           2) writes the variables SYSTEM_MODEL_param_RAMP.slx reads;
%           3) runs the model with logging of grid, DC-link and battery
%              signals, then plots input power vs output power and
%              calculates the efficiency of each stage.
%  Battery: the model's Battery block is a fixed 400 V / 250 Ah (100 kWh,
%           2C = 500 A) pack, so power is limited to 200 kW.
%% ======================================================================
clear; clc; close all;
modelName = 'SYSTEM_MODEL_param_RAMP_SOC';
pr  = @(name,val,unit,note) fprintf('  %-26s = %12.5g %-5s %s\n',name,val,unit,note);
sec = @(txt) fprintf('\n--- %s ---\n',txt);

%% 1. Inputs (press Enter for the default)
V_grid = input('Grid line-to-line RMS voltage [V]  (Enter = 400): ');
if isempty(V_grid), V_grid = 400; end
P_kW   = input('Required charging power [kW]        (Enter = 200): ');
if isempty(P_kW),   P_kW   = 200; end
assert(isscalar(V_grid) && V_grid>0 && isscalar(P_kW) && P_kW>0, 'Inputs must be positive
numbers.');
SOC0 = input('Initial battery SOC [%]              (Enter = 70): ');
if isempty(SOC0), SOC0 = 70; end
assert(isscalar(SOC0) && SOC0>=0 && SOC0<100, 'Initial SOC must be between 0 and 100 %.');
if SOC0 > 90
    warning('Initial SOC above 90 percent: the battery voltage is close to the 447 V CV target,
so the charging current will be limited.');
end
Vbat = 400;                                  % battery block nominal voltage [V]
if P_kW > 200
    warning('Battery pack in the model is 100 kWh / 2C = 200 kW max. Limiting power to 200
kW.');
    P_kW = 200;
end
P = P_kW*1e3;

%% 2. Fixed design assumptions (same as the original report)
Fs=50; Ws=2*pi*Fs; Fsw=10e3; Wsw=2*pi*Fsw; Ta=0.5/Fsw;
x=0.05; attenuation=0.05; R_per_H=37;
fsw=100e3; ripple_current=0.10; ripple_voltage=0.01; Vo_max=1.25*Vbat;

fprintf('\n=========== DESIGN CALCULATIONS: %g V grid, %g kW ===========\n',V_grid,P_kW);
sec('1. INPUTS AND ASSUMPTIONS');
pr('V_grid',V_grid,'V','grid line-to-line RMS (input)');
pr('P',P,'W','required charging power (input)');
pr('Vbat',Vbat,'V','battery nominal voltage (Battery block)');
pr('SOC0',SOC0,'%','initial battery state of charge (input, Battery block)');
pr('Fs',Fs,'Hz','grid frequency');
pr('Fsw',Fsw,'Hz','AFE switching frequency');
pr('fsw',fsw,'Hz','buck-boost switching frequency');
pr('I_line_rms',P/(sqrt(3)*V_grid),'A','P / (sqrt(3) V_grid)');

%% 3. AFE rectifier and LCL filter
sec('2. AFE RECTIFIER AND LCL FILTER');
Vdc_min = sqrt(2)*V_grid;
Vdc     = ceil(1.20*Vdc_min/50)*50;
Cdc     = P/(240*(0.1*Vdc)*(V_grid/sqrt(3))*Fs);
I_ripple= 0.2*sqrt(2)*P/(sqrt(3)*V_grid);
Zb=V_grid^2/P; Cb=1/(Ws*Zb);
```

```
L  = Vdc/(8*Fsw*I_ripple);
Cf = x*Cb;
a  = L*Cb*Wsw^2;
r  = (1-1/attenuation)/-(1-a*x);
if r<0, r=(1+1/attenuation)/-(1-a*x); end
Lf = L*r;  Lt = L+Lf;
Wres = sqrt(Lt/(L*Lf*Cf)); Fres = Wres/(2*pi);
rd_design = 1/(3*Wres*Cf);
R = R_per_H*L;  Rf = R_per_H*Lf;  Rt = R+Rf;
pr('Vdc_min',Vdc_min,'V','sqrt(2) V_grid');
pr('Vdc',Vdc,'V','ceil(1.2 Vdc_min / 50) * 50');
pr('Cdc',Cdc*1e6,'uF','P / (240 dV Vph Fs), dV = 10 % of Vdc');
pr('I_ripple',I_ripple,'A','0.2 sqrt(2) P / (sqrt(3) V_grid)');
pr('Zb',Zb,'Ohm','V_grid^2 / P');
pr('Cb',Cb*1e3,'mF','1 / (Ws Zb)');
pr('L',L*1e6,'uH','Vdc / (8 Fsw I_ripple)');
pr('Cf',Cf*1e6,'uF','x Cb, x = 0.05');
pr('a',a,'','L Cb Wsw^2');
pr('r',r,'','from attenuation factor 5 %');
pr('Lf',Lf*1e6,'uH','r L');
pr('Lt',Lt*1e6,'uH','L + Lf');
pr('Wres',Wres,'rad/s','sqrt(Lt / (L Lf Cf))');
pr('Fres',Fres/1e3,'kHz','Wres / (2 pi)');
pr('rd_design',rd_design,'Ohm','1 / (3 Wres Cf)  (model keeps 2 Ohm)');
pr('R',R,'Ohm','37 L');
pr('Rf',Rf,'Ohm','37 Lf');
pr('Rt',Rt,'Ohm','R + Rf');

%% 4. AFE controller gains
sec('3. AFE CONTROLLER');
fc  = min(Fsw/10, Fres/2.5);  wc=2*pi*fc;  wi=wc/10;
kpi = abs(1+1j*Ta*wc)*abs(Rt+1j*wc*Lt)/abs(1-1j*wi/wc);  kii = wi*kpi;
fc2 = 150;  wc2=2*pi*fc2;  wi2=wc2/5;
Kdc = 1.5*(V_grid/Vdc);
kpv = wc2*Cdc/(Kdc*abs(1-1j*wi2/wc2));  kiv = wi2*kpv;
Kmod = 1000/(Vdc/2);
m_index = sqrt(2/3)*V_grid/(Vdc/2);
pr('Ta',Ta*1e6,'us','1 / (2 Fsw)');
pr('fc (current loop)',fc,'Hz','min(Fsw/10, Fres/2.5)');
pr('wi (PI zero)',wi,'rad/s','wc / 10');
pr('kpi',kpi,'V/A','|1+jTa wc| |Rt+j wc Lt| / |1-j wi/wc|');
pr('kii',kii,'','wi kpi');
pr('fc2 (voltage loop)',fc2,'Hz','chosen');
pr('wi2 (PI zero)',wi2,'rad/s','wc2 / 5');
pr('Kdc',Kdc,'','1.5 V_grid / Vdc');
pr('kpv',kpv,'','wc2 Cdc / (Kdc |1-j wi2/wc2|)');
pr('kiv',kiv,'','wi2 kpv');
pr('Kmod',Kmod,'','1000 / (Vdc/2), carrier is +/-1000');
pr('m_index',m_index,'','sqrt(2/3) V_grid / (Vdc/2)');

%% 5. Synchronous buck-boost
sec('4. SYNCHRONOUS BUCK-BOOST');
Io_max = P/Vbat;
D_max  = Vo_max/(Vo_max+Vdc);
I_L    = Io_max/(1-D_max);
dIL    = ripple_current*I_L;
Lch_min= (Vdc*D_max)/(dIL*fsw);
Lch    = ceil(Lch_min/2e-6)*2e-6;
Co_min = (Io_max*D_max)/(fsw*ripple_voltage*Vo_max);
Co     = ceil(Co_min/100e-6)*100e-6;
IL_peak= I_L+0.5*dIL;
IL_SAT = 1.2*IL_peak;
Vsw_max= Vdc+Vo_max;
Dlim   = min(0.95, 1.13*D_max);
```

```
kpb    = 0.008     *(Lch/36e-6)*(1100/(Vdc+Vbat));
kib    = 3.0202337 *(Lch/36e-6)*(1100/(Vdc+Vbat));
Vcv    = 1.1175*Vbat;
pr('Vo_max',Vo_max,'V','1.25 Vbat');
pr('Io_max',Io_max,'A','P / Vbat');
pr('D_max',D_max,'','Vo_max / (Vo_max + Vdc)');
pr('I_L',I_L,'A','Io_max / (1 - D_max)');
pr('dIL',dIL,'A','10 % of I_L');
pr('Lch_min',Lch_min*1e6,'uH','Vdc D_max / (dIL fsw)');
pr('Lch',Lch*1e6,'uH','rounded up to 2 uH');
pr('Co_min',Co_min*1e6,'uF','Io_max D_max / (fsw dVo), dVo = 1 %');
pr('Co',Co*1e6,'uF','rounded up to 100 uF');
pr('IL_peak',IL_peak,'A','I_L + dIL/2');
pr('IL_SAT',IL_SAT,'A','1.2 IL_peak');
pr('Vsw_max',Vsw_max,'V','Vdc + Vo_max');
pr('Dlim',Dlim,'','min(0.95, 1.13 D_max)');
pr('kpb',kpb,'','0.008 (Lch/36u) (1100/(Vdc+Vbat))');
pr('kib',kib,'','3.02 (Lch/36u) (1100/(Vdc+Vbat))');
pr('Vcv',Vcv,'V','1.1175 Vbat (constant-voltage target)');

%% 6. Checks
sec('5. DESIGN CHECKS');
flags = {'** CHECK **','OK'};
fprintf('  6Fs < Fres < 0.5Fsw        : %s\n', flags{double(6*Fs<Fres && Fres<0.5*Fsw)+1});
fprintf('  Modulation index < 0.95    : %s (m = %.2f)\n', flags{double(m_index<0.95)+1},
m_index);
if Vsw_max > 0.75*1200 || IL_peak > 0.8*600
    fprintf('  NOTE: switch stress (%.0f V, %.0f A) exceeds a single 1200 V / 600 A module at
75/80 %% utilisation.\n',Vsw_max,IL_peak);
end
fprintf('\nWorkspace variables written for the model: V_grid SOC0 Vdc Cdc L Lf Cf R Rf Lt Fsw\
n');
fprintf('  kpi kii kpv kiv Kmod Io_max Lch Co Dlim kpb kib Vcv\n');

%% 7. Run the model and calculate power / efficiency
if ~exist([modelName '.slx'],'file')
    disp(['Put ' modelName '.slx in the current folder, then run this script again.']); return;
end
if strcmpi(input('Run the simulation now? (y/n, Enter = y): ','s'),'n'), return; end
T_sim = input('Simulation stop time [s]  (Enter = 0.5): ');
if isempty(T_sim), T_sim = 0.5; end

bdclose all; load_system(modelName);
sigs = {'Vabc','Iabc','Vdc','Idc','Vbuck','Ibuck','SOC'};
% remove logging blocks left in the model by an earlier run (if any)
old = find_system(modelName,'SearchDepth',1,'regexp','on','Name','^LOG_');
for k = 1:numel(old)
    try
        lh = get_param(old{k},'LineHandles');
        fn = fieldnames(lh);
        for q = 1:numel(fn)
            hh = lh.(fn{q});
            for z = 1:numel(hh)
                if hh(z) > 0 && ishandle(hh(z)), delete_line(hh(z)); end
            end
        end
        delete_block(old{k});
    catch
    end
end
try
    for k = 1:numel(sigs)
        y = 80 + 50*k;
        add_block('simulink/Signal Routing/From',[modelName '/LOG_From_' sigs{k}], ...
```

```
            'GotoTag',sigs{k},'Position',[1000 y 1060 y+20]);
        add_block('simulink/Sinks/To Workspace',[modelName '/LOG_' sigs{k}], ...
            'VariableName',['log_' sigs{k}],'SaveFormat','Timeseries', ...
            'MaxDataPoints','inf','Position',[1120 y 1200 y+20]);
        add_line(modelName,['LOG_From_' sigs{k} '/1'],['LOG_' sigs{k} '/1']);
    end
    set_param(modelName,'StopTime',num2str(T_sim));
    open_system(modelName);
    simOut = sim(modelName,'ReturnWorkspaceOutputs','on');
catch ME
    disp('Simulation or logging failed:'); disp(ME.message); return;
end
disp('Simulation done. Calculating power and efficiency...');

dtu = 20e-6;                                % uniform analysis step
ts0 = simOut.get('log_Vabc');
t0  = ts0.Time(:);
tu  = (min(t0):dtu:max(t0))';
D   = struct();
for k = 1:numel(sigs)
    ts = simOut.get(['log_' sigs{k}]);
    d  = squeeze(ts.Data);
    if size(d,1) ~= numel(ts.Time), d = d.'; end
    [tk,ik] = unique(ts.Time(:));
    D.(sigs{k}) = interp1(tk, d(ik,:), tu, 'linear', 'extrap');
end
Nw   = round(0.02/dtu);                     % one grid period moving average
Pin  = movmean(sum(D.Vabc.*D.Iabc,2), Nw);  % grid input power
Pdc  = movmean(D.Vdc.*D.Idc,          Nw);  % power leaving the AFE bridge
Pout = movmean(D.Vbuck.*D.Ibuck,      Nw);  % power into the battery

if T_sim >= 0.4, w1 = 0.25; w2 = 0.38; else, w1 = 0.6*T_sim; w2 = T_sim; end
idx = tu >= w1 & tu <= w2;
Pin_w = mean(Pin(idx)); Pdc_w = mean(Pdc(idx)); Pout_w = mean(Pout(idx));
eta_afe = Pdc_w/Pin_w; eta_dcdc = Pout_w/Pdc_w; eta_tot = Pout_w/Pin_w;
idx2 = tu >= 0.05;
eta_energy = trapz(tu(idx2),Pout(idx2))/trapz(tu(idx2),Pin(idx2));

fprintf('\n=========== POWER AND EFFICIENCY (window %.2f-%.2f s) ===========\n',w1,w2);
pr('Input power (grid)',Pin_w/1e3,'kW','mean of v.i over the window');
pr('DC-link power',Pdc_w/1e3,'kW','Vdc . Idc at the rectifier output');
pr('Output power (battery)',Pout_w/1e3,'kW','Vbuck . Ibuck');
pr('AFE + LCL efficiency',100*eta_afe,'%','Pdc / Pin');
pr('Buck-boost efficiency',100*eta_dcdc,'%','Pout / Pdc');
pr('Overall efficiency',100*eta_tot,'%','Pout / Pin');
pr('AFE + LCL loss',(Pin_w-Pdc_w)/1e3,'kW','Pin - Pdc');
pr('Buck-boost loss',(Pdc_w-Pout_w)/1e3,'kW','Pdc - Pout');
pr('Energy efficiency (>0.05 s)',100*eta_energy,'%','integral Pout / integral Pin');
fprintf('  Note: the efficiency covers the simulated components (switch and diode drops,\n
filter and inductor resistance, damping branch). Battery internal loss is excluded.\n');
%% Grid-side power factor over a whole number of grid periods in the same window
nper = floor((w2-w1)*Fs);  if nper < 1, nper = 1; end
ip   = tu >= w1 & tu < w1 + nper/Fs;
tt   = tu(ip);  vv = D.Vabc(ip,:);  ia = D.Iabc(ip,:);
P3   = mean(sum(vv.*ia,2));                 % real power [W]
Vrms = sqrt(mean(vv.^2,1));                 % rms of each phase voltage
Irms = sqrt(mean(ia.^2,1));                 % rms of each phase current
S3   = sum(Vrms.*Irms);                     % apparent power [VA]
PF   = P3/S3;                               % true power factor (includes distortion)
ee   = exp(-1j*2*pi*Fs*tt);
V1   = 2*mean(vv.*ee,1);                    % fundamental phasors (peak) per phase
I1   = 2*mean(ia.*ee,1);
phi  = mean(angle(V1.*conj(I1)));           % > 0 means current lags voltage
DPF  = cos(phi);                            % displacement power factor
```

```
I1rms= abs(I1)/sqrt(2);
THDi = 100*mean(sqrt(max(Irms.^2 - I1rms.^2,0))./I1rms);
fprintf('\n========== GRID-SIDE POWER FACTOR (window %.2f-%.2f s, %d grid periods) ==========\
n',w1,w1+nper/Fs,nper);
pr('Real power P',P3/1e3,'kW','mean of v.i');
pr('Apparent power S',S3/1e3,'kVA','sum of Vrms . Irms over the 3 phases');
pr('True power factor',PF,'','P / S');
pr('Displacement power factor',DPF,'','cos(phase angle of the 50 Hz components)');
pr('Voltage-current angle',phi*180/pi,'deg','> 0 means current lags voltage');
pr('Fundamental current (rms)',mean(I1rms),'A','50 Hz component, mean of 3 phases');
pr('Current distortion (THD)',THDi,'%','includes the switching ripple');
fprintf('\n========== BATTERY STATE OF CHARGE ==========\n');
pr('SOC at start',D.SOC(1),'%','from the Battery block');
pr('SOC at end',D.SOC(end),'%','at the stop time');
pr('SOC change',D.SOC(end)-D.SOC(1),'%','end minus start');
eff = struct('window',[w1 w2],'Pin_kW',Pin_w/1e3,'Pdc_kW',Pdc_w/1e3,'Pout_kW',Pout_w/1e3, ...

'eta_afe',eta_afe,'eta_dcdc',eta_dcdc,'eta_total',eta_tot,'eta_energy',eta_energy,'PF',PF,'DPF',
DPF); %#ok<NASGU>

fig = figure('Name','Input vs output power','Color','w','Position',[80 60 1000 820]);
subplot(3,1,1);
plot(tu,Pin/1e3,'b',tu,Pdc/1e3,'g',tu,Pout/1e3,'r','LineWidth',1.5); grid on;
legend('P_{in} (grid)','P_{DC link}','P_{out} (battery)','Location','best');
ylabel('Power [kW]'); title(sprintf('Input vs output power (%g V, %g kW), 20 ms moving
average',V_grid,P_kW));
subplot(3,1,2);
plot(tu,(Pin-Pdc)/1e3,'g',tu,(Pdc-Pout)/1e3,'r',tu,(Pin-Pout)/1e3,'k','LineWidth',1.3); grid on;
legend('AFE + LCL loss','Buck-boost loss','Total loss','Location','best'); ylabel('Loss [kW]');
subplot(3,1,3);
mask = Pin > 0.05*P;
e1 = 100*Pdc./Pin;  e1(~mask) = NaN;
e2 = 100*Pout./Pdc; e2(~mask) = NaN;
e3 = 100*Pout./Pin; e3(~mask) = NaN;
plot(tu,e1,'g',tu,e2,'r',tu,e3,'k','LineWidth',1.3); grid on; ylim([60 110]);
legend('AFE + LCL','Buck-boost','Overall','Location','best');
ylabel('Efficiency [%]'); xlabel('Time [s]');
title('Efficiency vs time (values above 100 % are transients: stored inductor and capacitor
energy)');
try, exportgraphics(fig,'power_efficiency.png','Resolution',200); catch,
saveas(fig,'power_efficiency.png'); end
disp('Saved plot: power_efficiency.png');
disp('Also check the model scopes: the SOC should rise above its initial value.');
```

Appendix B: Command Window Output for the Four Cases

Case 1: 400 V, 200 kW, initial SOC 80 %, 5 s

```
Grid line-to-line RMS voltage [V]  (Enter = 400): 400
Required charging power [kW]        (Enter = 200): 200
Initial battery SOC [%]              (Enter = 70): 80

========== DESIGN CALCULATIONS: 400 V grid, 200 kW ==========

--- 1. INPUTS AND ASSUMPTIONS ---
  V_grid                     =          400 V     grid line-to-line RMS (input)
  P                          =        2e+05 W     required charging power (input)
  Vbat                       =          400 V     battery nominal voltage (Battery block)
  SOC0                       =           80 %     initial battery state of charge (input,
Battery block)
  Fs                         =           50 Hz    grid frequency
  Fsw                        =        10000 Hz    AFE switching frequency
  fsw                        =        1e+05 Hz    buck-boost switching frequency
  I_line_rms                 =       288.68 A     P / (sqrt(3) V_grid)

--- 2. AFE RECTIFIER AND LCL FILTER ---
  Vdc_min                    =       565.69 V     sqrt(2) V_grid
  Vdc                        =          700 V     ceil(1.2 Vdc_min / 50) * 50
  Cdc                        =         1031 uF    P / (240 dV Vph Fs), dV = 10 % of Vdc
  I_ripple                   =        81.65 A     0.2 sqrt(2) P / (sqrt(3) V_grid)
  Zb                         =          0.8 Ohm   V_grid^2 / P
  Cb                         =       3.9789 mF    1 / (Ws Zb)
  L                          =       107.17 uH    Vdc / (8 Fsw I_ripple)
  Cf                         =       198.94 uF    x Cb, x = 0.05
  a                          =       1683.3       L Cb Wsw^2
  r                          =       0.2525       from attenuation factor 5 %
  Lf                         =        27.06 uH    r L
  Lt                         =       134.22 uH    L + Lf
  Wres                       =        15253 rad/s sqrt(Lt / (L Lf Cf))
  Fres                       =       2.4276 kHz   Wres / (2 pi)
  rd_design                  =      0.10985 Ohm   1 / (3 Wres Cf)  (model keeps 2 Ohm)
  R                          =    0.0039651 Ohm   37 L
  Rf                         =    0.0010012 Ohm   37 Lf
  Rt                         =    0.0049663 Ohm   R + Rf

--- 3. AFE CONTROLLER ---
  Ta                         =           50 us    1 / (2 Fsw)
  fc (current loop)          =       971.06 Hz    min(Fsw/10, Fres/2.5)
  wi (PI zero)               =       610.13 rad/s wc / 10
  kpi                        =      0.85198 V/A   |1+jTa wc| |Rt+j wc Lt| / |1-j wi/wc|
  kii                        =       519.82       wi kpi
  fc2 (voltage loop)         =          150 Hz    chosen
  wi2 (PI zero)              =        188.5 rad/s wc2 / 5
  Kdc                        =      0.85714       1.5 V_grid / Vdc
  kpv                        =       1.1116       wc2 Cdc / (Kdc |1-j wi2/wc2|)
  kiv                        =       209.53       wi2 kpv
  Kmod                       =       2.8571       1000 / (Vdc/2), carrier is +/-1000
  m_index                    =      0.93314       sqrt(2/3) V_grid / (Vdc/2)

--- 4. SYNCHRONOUS BUCK-BOOST ---
  Vo_max                     =          500 V     1.25 Vbat
  Io_max                     =          500 A     P / Vbat
  D_max                      =      0.41667       Vo_max / (Vo_max + Vdc)
```

```
  I_L                       =       857.14 A     Io_max / (1 - D_max)
  dIL                       =       85.714 A     10 % of I_L
  Lch_min                   =       34.028 uH    Vdc D_max / (dIL fsw)
  Lch                       =           36 uH    rounded up to 2 uH
  Co_min                    =       416.67 uF    Io_max D_max / (fsw dVo), dVo = 1 %
  Co                        =          500 uF    rounded up to 100 uF
  IL_peak                   =          900 A     I_L + dIL/2
  IL_SAT                    =         1080 A     1.2 IL_peak
  Vsw_max                   =         1200 V     Vdc + Vo_max
  Dlim                      =      0.47083       min(0.95, 1.13 D_max)
  kpb                       =        0.008       0.008 (Lch/36u) (1100/(Vdc+Vbat))
  kib                       =       3.0202       3.02 (Lch/36u) (1100/(Vdc+Vbat))
  Vcv                       =          447 V     1.1175 Vbat (constant-voltage target)

--- 5. DESIGN CHECKS ---
  6Fs < Fres < 0.5Fsw       : OK
  Modulation index < 0.95   : OK (m = 0.93)
  NOTE: switch stress (1200 V, 900 A) exceeds a single 1200 V / 600 A module at 75/80 %
utilisation.

Workspace variables written for the model: V_grid SOC0 Vdc Cdc L Lf Cf R Rf Lt Fsw
  kpi kii kpv kiv Kmod Io_max Lch Co Dlim kpb kib Vcv
Run the simulation now? (y/n, Enter = y):
Simulation stop time [s]  (Enter = 0.5): 5

Simulation done. Calculating power and efficiency...

=========== POWER AND EFFICIENCY (window 0.25-0.38 s) ===========
  Input power (grid)        =       250.16 kW    mean of v.i over the window
  DC-link power             =       249.15 kW    Vdc . Idc at the rectifier output
  Output power (battery)    =       225.07 kW    Vbuck . Ibuck
  AFE + LCL efficiency      =       99.596 %     Pdc / Pin
  Buck-boost efficiency     =       90.337 %     Pout / Pdc
  Overall efficiency        =       89.972 %     Pout / Pin
  AFE + LCL loss            =       1.0095 kW    Pin - Pdc
  Buck-boost loss           =       24.076 kW    Pdc - Pout
  Energy efficiency (>0.05 s) =       89.862 %     integral Pout / integral Pin
  Note: the efficiency covers the simulated components (switch and diode drops,
  filter and inductor resistance, damping branch). Battery internal loss is excluded.
=========== GRID-SIDE POWER FACTOR (window 0.25-0.37 s, 6 grid periods) ===========
  Real power P              =       250.06 kW    mean of v.i
  Apparent power S          =       250.29 kVA   sum of Vrms . Irms over the 3 phases
  True power factor         =      0.99906       P / S
  Displacement power factor =            1       cos(phase angle of the 50 Hz components)
  Voltage-current angle     =    -0.022284 deg   > 0 means current lags voltage
  Fundamental current (rms) =       360.92 A     50 Hz component, mean of 3 phases
  Current distortion (THD)  =       4.3432 %     includes the switching ripple
=========== BATTERY STATE OF CHARGE ===========
  SOC at start              =           80 %     from the Battery block
  SOC at end                =       80.134 %     at the stop time
  SOC change                =      0.13351 %     end minus start
Saved plot: power_efficiency.png
```

Case 2: 400 V, 200 kW, initial SOC 70 %, 10 s

```
Grid line-to-line RMS voltage [V]  (Enter = 400): 400
Required charging power [kW]        (Enter = 200): 200
Initial battery SOC [%]              (Enter = 70): 70

=========== DESIGN CALCULATIONS: 400 V grid, 200 kW ===========

--- 1. INPUTS AND ASSUMPTIONS ---
  V_grid                     =          400 V     grid line-to-line RMS (input)
  P                          =        2e+05 W     required charging power (input)
  Vbat                       =          400 V     battery nominal voltage (Battery block)
  SOC0                       =           70 %     initial battery state of charge (input,
Battery block)
  Fs                         =           50 Hz    grid frequency
  Fsw                        =        10000 Hz    AFE switching frequency
  fsw                        =        1e+05 Hz    buck-boost switching frequency
  I_line_rms                 =       288.68 A     P / (sqrt(3) V_grid)

--- 2. AFE RECTIFIER AND LCL FILTER ---
  Vdc_min                    =       565.69 V     sqrt(2) V_grid
  Vdc                        =          700 V     ceil(1.2 Vdc_min / 50) * 50
  Cdc                        =         1031 uF    P / (240 dV Vph Fs), dV = 10 % of Vdc
  I_ripple                   =        81.65 A     0.2 sqrt(2) P / (sqrt(3) V_grid)
  Zb                         =          0.8 Ohm   V_grid^2 / P
  Cb                         =       3.9789 mF    1 / (Ws Zb)
  L                          =       107.17 uH    Vdc / (8 Fsw I_ripple)
  Cf                         =       198.94 uF    x Cb, x = 0.05
  a                          =       1683.3       L Cb Wsw^2
  r                          =       0.2525       from attenuation factor 5 %
  Lf                         =        27.06 uH    r L
  Lt                         =       134.22 uH    L + Lf
  Wres                       =        15253 rad/s sqrt(Lt / (L Lf Cf))
  Fres                       =       2.4276 kHz   Wres / (2 pi)
  rd_design                  =      0.10985 Ohm   1 / (3 Wres Cf)  (model keeps 2 Ohm)
  R                          =    0.0039651 Ohm   37 L
  Rf                         =    0.0010012 Ohm   37 Lf
  Rt                         =    0.0049663 Ohm   R + Rf

--- 3. AFE CONTROLLER ---
  Ta                         =           50 us    1 / (2 Fsw)
  fc (current loop)          =       971.06 Hz    min(Fsw/10, Fres/2.5)
  wi (PI zero)               =       610.13 rad/s wc / 10
  kpi                        =      0.85198 V/A   |1+jTa wc| |Rt+j wc Lt| / |1-j wi/wc|
  kii                        =       519.82       wi kpi
  fc2 (voltage loop)         =          150 Hz    chosen
  wi2 (PI zero)              =        188.5 rad/s wc2 / 5
  Kdc                        =      0.85714       1.5 V_grid / Vdc
  kpv                        =       1.1116       wc2 Cdc / (Kdc |1-j wi2/wc2|)
  kiv                        =       209.53       wi2 kpv
  Kmod                       =       2.8571       1000 / (Vdc/2), carrier is +/-1000
  m_index                    =      0.93314       sqrt(2/3) V_grid / (Vdc/2)

--- 4. SYNCHRONOUS BUCK-BOOST ---
  Vo_max                     =          500 V     1.25 Vbat
  Io_max                     =          500 A     P / Vbat
  D_max                      =      0.41667       Vo_max / (Vo_max + Vdc)
  I_L                        =       857.14 A     Io_max / (1 - D_max)
  dIL                        =       85.714 A     10 % of I_L
```

```
  Lch_min                    =       34.028 uH    Vdc D_max / (dIL fsw)
  Lch                        =           36 uH    rounded up to 2 uH
  Co_min                     =       416.67 uF    Io_max D_max / (fsw dVo), dVo = 1 %
  Co                         =          500 uF    rounded up to 100 uF
  IL_peak                    =          900 A     I_L + dIL/2
  IL_SAT                     =         1080 A     1.2 IL_peak
  Vsw_max                    =         1200 V     Vdc + Vo_max
  Dlim                       =      0.47083       min(0.95, 1.13 D_max)
  kpb                        =        0.008       0.008 (Lch/36u) (1100/(Vdc+Vbat))
  kib                        =       3.0202       3.02 (Lch/36u) (1100/(Vdc+Vbat))
  Vcv                        =          447 V     1.1175 Vbat (constant-voltage target)

--- 5. DESIGN CHECKS ---
  6Fs < Fres < 0.5Fsw        : OK
  Modulation index < 0.95    : OK (m = 0.93)
  NOTE: switch stress (1200 V, 900 A) exceeds a single 1200 V / 600 A module at 75/80 %
utilisation.

Workspace variables written for the model: V_grid SOC0 Vdc Cdc L Lf Cf R Rf Lt Fsw
  kpi kii kpv kiv Kmod Io_max Lch Co Dlim kpb kib Vcv
Run the simulation now? (y/n, Enter = y):
Simulation stop time [s]  (Enter = 0.5): 10

Simulation done. Calculating power and efficiency...

=========== POWER AND EFFICIENCY (window 0.25-0.38 s) ===========
  Input power (grid)         =       248.57 kW    mean of v.i over the window
  DC-link power              =       246.48 kW    Vdc . Idc at the rectifier output
  Output power (battery)     =       223.64 kW    Vbuck . Ibuck
  AFE + LCL efficiency       =       99.159 %     Pdc / Pin
  Buck-boost efficiency      =       90.734 %     Pout / Pdc
  Overall efficiency         =       89.971 %     Pout / Pin
  AFE + LCL loss             =       2.0903 kW    Pin - Pdc
  Buck-boost loss            =       22.839 kW    Pdc - Pout
  Energy efficiency (>0.05 s) =       90.376 %     integral Pout / integral Pin
  Note: the efficiency covers the simulated components (switch and diode drops,
  filter and inductor resistance, damping branch). Battery internal loss is excluded.
=========== GRID-SIDE POWER FACTOR (window 0.25-0.37 s, 6 grid periods) ===========
  Real power P               =       248.44 kW    mean of v.i
  Apparent power S           =       248.67 kVA   sum of Vrms . Irms over the 3 phases
  True power factor          =      0.99907       P / S
  Displacement power factor  =            1       cos(phase angle of the 50 Hz components)
  Voltage-current angle      =    -0.020112 deg   > 0 means current lags voltage
  Fundamental current (rms)  =       358.59 A     50 Hz component, mean of 3 phases
  Current distortion (THD)   =       4.3105 %     includes the switching ripple
=========== BATTERY STATE OF CHARGE ===========
  SOC at start               =           70 %     from the Battery block
  SOC at end                 =       70.337 %     at the stop time
  SOC change                 =      0.33703 %     end minus start
Saved plot: power_efficiency.png
```

Case 3: 400 V, 200 kW, initial SOC 90 %, 3 s

```
Grid line-to-line RMS voltage [V]  (Enter = 400): 400
Required charging power [kW]        (Enter = 200): 200
Initial battery SOC [%]              (Enter = 70): 90

========== DESIGN CALCULATIONS: 400 V grid, 200 kW ==========

--- 1. INPUTS AND ASSUMPTIONS ---
  V_grid                    =          400 V     grid line-to-line RMS (input)
  P                         =        2e+05 W     required charging power (input)
  Vbat                      =          400 V     battery nominal voltage (Battery block)
  SOC0                      =           90 %     initial battery state of charge (input,
Battery block)
  Fs                        =           50 Hz    grid frequency
  Fsw                       =        10000 Hz    AFE switching frequency
  fsw                       =        1e+05 Hz    buck-boost switching frequency
  I_line_rms                =       288.68 A     P / (sqrt(3) V_grid)

--- 2. AFE RECTIFIER AND LCL FILTER ---
  Vdc_min                   =       565.69 V     sqrt(2) V_grid
  Vdc                       =          700 V     ceil(1.2 Vdc_min / 50) * 50
  Cdc                       =         1031 uF    P / (240 dV Vph Fs), dV = 10 % of Vdc
  I_ripple                  =        81.65 A     0.2 sqrt(2) P / (sqrt(3) V_grid)
  Zb                        =          0.8 Ohm   V_grid^2 / P
  Cb                        =       3.9789 mF    1 / (Ws Zb)
  L                         =       107.17 uH    Vdc / (8 Fsw I_ripple)
  Cf                        =       198.94 uF    x Cb, x = 0.05
  a                         =       1683.3       L Cb Wsw^2
  r                         =       0.2525       from attenuation factor 5 %
  Lf                        =        27.06 uH    r L
  Lt                        =       134.22 uH    L + Lf
  Wres                      =        15253 rad/s sqrt(Lt / (L Lf Cf))
  Fres                      =       2.4276 kHz   Wres / (2 pi)
  rd_design                 =      0.10985 Ohm   1 / (3 Wres Cf)  (model keeps 2 Ohm)
  R                         =    0.0039651 Ohm   37 L
  Rf                        =    0.0010012 Ohm   37 Lf
  Rt                        =    0.0049663 Ohm   R + Rf

--- 3. AFE CONTROLLER ---
  Ta                        =           50 us    1 / (2 Fsw)
  fc (current loop)         =       971.06 Hz    min(Fsw/10, Fres/2.5)
  wi (PI zero)              =       610.13 rad/s wc / 10
  kpi                       =      0.85198 V/A   |1+jTa wc| |Rt+j wc Lt| / |1-j wi/wc|
  kii                       =       519.82       wi kpi
  fc2 (voltage loop)        =          150 Hz    chosen
  wi2 (PI zero)             =        188.5 rad/s wc2 / 5
  Kdc                       =      0.85714       1.5 V_grid / Vdc
  kpv                       =       1.1116       wc2 Cdc / (Kdc |1-j wi2/wc2|)
  kiv                       =       209.53       wi2 kpv
  Kmod                      =       2.8571       1000 / (Vdc/2), carrier is +/-1000
  m_index                   =      0.93314       sqrt(2/3) V_grid / (Vdc/2)

--- 4. SYNCHRONOUS BUCK-BOOST ---
  Vo_max                    =          500 V     1.25 Vbat
  Io_max                    =          500 A     P / Vbat
  D_max                     =      0.41667       Vo_max / (Vo_max + Vdc)
  I_L                       =       857.14 A     Io_max / (1 - D_max)
  dIL                       =       85.714 A     10 % of I_L
```

```
  Lch_min                    =       34.028 uH    Vdc D_max / (dIL fsw)
  Lch                        =           36 uH    rounded up to 2 uH
  Co_min                     =       416.67 uF    Io_max D_max / (fsw dVo), dVo = 1 %
  Co                         =          500 uF    rounded up to 100 uF
  IL_peak                    =          900 A     I_L + dIL/2
  IL_SAT                     =         1080 A     1.2 IL_peak
  Vsw_max                    =         1200 V     Vdc + Vo_max
  Dlim                       =      0.47083       min(0.95, 1.13 D_max)
  kpb                        =        0.008       0.008 (Lch/36u) (1100/(Vdc+Vbat))
  kib                        =       3.0202       3.02 (Lch/36u) (1100/(Vdc+Vbat))
  Vcv                        =          447 V     1.1175 Vbat (constant-voltage target)

--- 5. DESIGN CHECKS ---
  6Fs < Fres < 0.5Fsw        : OK
  Modulation index < 0.95    : OK (m = 0.93)
  NOTE: switch stress (1200 V, 900 A) exceeds a single 1200 V / 600 A module at 75/80 %
utilisation.

Workspace variables written for the model: V_grid SOC0 Vdc Cdc L Lf Cf R Rf Lt Fsw
  kpi kii kpv kiv Kmod Io_max Lch Co Dlim kpb kib Vcv
Run the simulation now? (y/n, Enter = y):
Simulation stop time [s]  (Enter = 0.5): 3

Simulation done. Calculating power and efficiency...

=========== POWER AND EFFICIENCY (window 0.25-0.38 s) ===========
  Input power (grid)         =       252.88 kW    mean of v.i over the window
  DC-link power              =        251.5 kW    Vdc . Idc at the rectifier output
  Output power (battery)     =        227.6 kW    Vbuck . Ibuck
  AFE + LCL efficiency       =       99.453 %     Pdc / Pin
  Buck-boost efficiency      =       90.497 %     Pout / Pdc
  Overall efficiency         =       90.002 %     Pout / Pin
  AFE + LCL loss             =       1.3838 kW    Pin - Pdc
  Buck-boost loss            =         23.9 kW    Pdc - Pout
  Energy efficiency (>0.05 s) =       88.691 %     integral Pout / integral Pin
  Note: the efficiency covers the simulated components (switch and diode drops,
  filter and inductor resistance, damping branch). Battery internal loss is excluded.
=========== GRID-SIDE POWER FACTOR (window 0.25-0.37 s, 6 grid periods) ===========
  Real power P               =       252.72 kW    mean of v.i
  Apparent power S           =       252.95 kVA   sum of Vrms . Irms over the 3 phases
  True power factor          =       0.9991       P / S
  Displacement power factor  =            1       cos(phase angle of the 50 Hz components)
  Voltage-current angle      =    -0.018855 deg   > 0 means current lags voltage
  Fundamental current (rms)  =       364.77 A     50 Hz component, mean of 3 phases
  Current distortion (THD)   =       4.2441 %     includes the switching ripple
=========== BATTERY STATE OF CHARGE ===========
  SOC at start               =           90 %     from the Battery block
  SOC at end                 =        90.06 %     at the stop time
  SOC change                 =     0.060404 %     end minus start
Saved plot: power_efficiency.png
```

Case 4: 220 V, 60 kW, initial SOC 70 %, 0.5 s

```
Grid line-to-line RMS voltage [V]  (Enter = 400): 220
Required charging power [kW]        (Enter = 200): 60
Initial battery SOC [%]              (Enter = 70): 70

=========== DESIGN CALCULATIONS: 220 V grid, 60 kW ===========

--- 1. INPUTS AND ASSUMPTIONS ---
  V_grid                     =          220 V     grid line-to-line RMS (input)
  P                          =        60000 W     required charging power (input)
  Vbat                       =          400 V     battery nominal voltage (Battery block)
  SOC0                       =           70 %     initial battery state of charge (input,
Battery block)
  Fs                         =           50 Hz    grid frequency
  Fsw                        =        10000 Hz    AFE switching frequency
  fsw                        =        1e+05 Hz    buck-boost switching frequency
  I_line_rms                 =       157.46 A     P / (sqrt(3) V_grid)

--- 2. AFE RECTIFIER AND LCL FILTER ---
  Vdc_min                    =       311.13 V     sqrt(2) V_grid
  Vdc                        =          400 V     ceil(1.2 Vdc_min / 50) * 50
  Cdc                        =       984.12 uF    P / (240 dV Vph Fs), dV = 10 % of Vdc
  I_ripple                   =       44.536 A     0.2 sqrt(2) P / (sqrt(3) V_grid)
  Zb                         =      0.80667 Ohm   V_grid^2 / P
  Cb                         =        3.946 mF    1 / (Ws Zb)
  L                          =       112.27 uH    Vdc / (8 Fsw I_ripple)
  Cf                         =        197.3 uF    x Cb, x = 0.05
  a                          =       1748.9       L Cb Wsw^2
  r                          =      0.24292       from attenuation factor 5 %
  Lf                         =       27.273 uH    r L
  Lt                         =       139.54 uH    L + Lf
  Wres                       =        15198 rad/s sqrt(Lt / (L Lf Cf))
  Fres                       =       2.4189 kHz   Wres / (2 pi)
  rd_design                  =      0.11116 Ohm   1 / (3 Wres Cf)  (model keeps 2 Ohm)
  R                          =    0.0041539 Ohm   37 L
  Rf                         =    0.0010091 Ohm   37 Lf
  Rt                         =     0.005163 Ohm   R + Rf

--- 3. AFE CONTROLLER ---
  Ta                         =           50 us    1 / (2 Fsw)
  fc (current loop)          =       967.55 Hz    min(Fsw/10, Fres/2.5)
  wi (PI zero)               =       607.93 rad/s wc / 10
  kpi                        =      0.88226 V/A   |1+jTa wc| |Rt+j wc Lt| / |1-j wi/wc|
  kii                        =       536.35       wi kpi
  fc2 (voltage loop)         =          150 Hz    chosen
  wi2 (PI zero)              =        188.5 rad/s wc2 / 5
  Kdc                        =        0.825       1.5 V_grid / Vdc
  kpv                        =       1.1024       wc2 Cdc / (Kdc |1-j wi2/wc2|)
  kiv                        =        207.8       wi2 kpv
  Kmod                       =            5       1000 / (Vdc/2), carrier is +/-1000
  m_index                    =      0.89815       sqrt(2/3) V_grid / (Vdc/2)

--- 4. SYNCHRONOUS BUCK-BOOST ---
  Vo_max                     =          500 V     1.25 Vbat
  Io_max                     =          150 A     P / Vbat
  D_max                      =      0.55556       Vo_max / (Vo_max + Vdc)
  I_L                        =        337.5 A     Io_max / (1 - D_max)
  dIL                        =        33.75 A     10 % of I_L
```

```
  Lch_min                    =       65.844 uH    Vdc D_max / (dIL fsw)
  Lch                        =           66 uH    rounded up to 2 uH
  Co_min                     =       166.67 uF    Io_max D_max / (fsw dVo), dVo = 1 %
  Co                         =          200 uF    rounded up to 100 uF
  IL_peak                    =       354.38 A     I_L + dIL/2
  IL_SAT                     =       425.25 A     1.2 IL_peak
  Vsw_max                    =          900 V     Vdc + Vo_max
  Dlim                       =      0.62778       min(0.95, 1.13 D_max)
  kpb                        =     0.020167       0.008 (Lch/36u) (1100/(Vdc+Vbat))
  kib                        =       7.6135       3.02 (Lch/36u) (1100/(Vdc+Vbat))
  Vcv                        =          447 V     1.1175 Vbat (constant-voltage target)

--- 5. DESIGN CHECKS ---
  6Fs < Fres < 0.5Fsw        : OK
  Modulation index < 0.95    : OK (m = 0.90)

Workspace variables written for the model: V_grid SOC0 Vdc Cdc L Lf Cf R Rf Lt Fsw
  kpi kii kpv kiv Kmod Io_max Lch Co Dlim kpb kib Vcv
Run the simulation now? (y/n, Enter = y): y
Simulation stop time [s]  (Enter = 0.5): 0.5

Simulation done. Calculating power and efficiency...

=========== POWER AND EFFICIENCY (window 0.25-0.38 s) ===========
  Input power (grid)         =       74.443 kW    mean of v.i over the window
  DC-link power              =       73.361 kW    Vdc . Idc at the rectifier output
  Output power (battery)     =       68.988 kW    Vbuck . Ibuck
  AFE + LCL efficiency       =       98.547 %     Pdc / Pin
  Buck-boost efficiency      =       94.039 %     Pout / Pdc
  Overall efficiency         =       92.672 %     Pout / Pin
  AFE + LCL loss             =        1.082 kW    Pin - Pdc
  Buck-boost loss            =       4.3733 kW    Pdc - Pout
  Energy efficiency (>0.05 s) =       92.497 %     integral Pout / integral Pin
  Note: the efficiency covers the simulated components (switch and diode drops,
  filter and inductor resistance, damping branch). Battery internal loss is excluded.
=========== GRID-SIDE POWER FACTOR (window 0.25-0.37 s, 6 grid periods) ===========
  Real power P               =       74.289 kW    mean of v.i
  Apparent power S           =       75.175 kVA   sum of Vrms . Irms over the 3 phases
  True power factor          =      0.98822       P / S
  Displacement power factor  =            1       cos(phase angle of the 50 Hz components)
  Voltage-current angle      =    -0.025768 deg   > 0 means current lags voltage
  Fundamental current (rms)  =       194.96 A     50 Hz component, mean of 3 phases
  Current distortion (THD)   =       15.486 %     includes the switching ripple
=========== BATTERY STATE OF CHARGE ===========
  SOC at start               =           70 %     from the Battery block
  SOC at end                 =       70.007 %     at the stop time
  SOC change                 =    0.0071273 %     end minus start
Saved plot: power_efficiency.png
```

Appendix C: Challenges Encountered and How They Were Solved

The work went through two phases. In Phase 1 the aim was to generalise the design script and repair an earlier version of the Simulink model (PFC_2). Many problems were found and fixed, but the DC-link voltage still did not reach its target and the battery SOC did not increase. In Phase 2 the project was restarted from the original Simulink model, which was made parametric, tuned and tested. The two tables below list the difficulties and how each was handled.

C.1 Phase 1: generalising the script and repairing the earlier model

Difficulty	What was observed / cause	How it was solved
Stale workspace variables	Simulations ran against old or missing variables because the design script had not been run just before.	Run bdclose all, clear mex and the design script before every simulation; the final script also runs the model itself.
Simulink reusing an open model	An edited file looked unchanged because a model with the same name was already open in memory.	Every revision is saved under a new file name (for example _param and _RAMP).
Clipped scope axis	A fixed ±2000 A axis hid the true peak currents, so different runs looked identical.	Autoscale every scope before interpreting a plot.
Design values not reaching the model	Cdc, the buck-boost inductor and the buck-boost capacitor were typed numbers, so changing the inputs changed nothing.	Replaced the numbers with workspace variables; extended to the grid voltage, current references, limits and gains in the final model.
LCL filter built wrongly	Both filter inductors used the same variable, so the filter was symmetric, not the designed LCL.	Gave the converter side and grid side their own inductance and resistance variables.
Solver and frequency settings	A variable-step solver was combined with a discrete powergui, and the powergui frequency was still 60 Hz.	Fixed-step settings and a 50 Hz frequency were used in the earlier model. The final model returns to the original settings (ode45 variable-step with a 3 µs discrete powergui, 50 Hz), which ran without errors.
Comparator operator	The PWM comparators had no explicit operator, so the default equality test left the gates almost always off.	Set the operator to greater-than or equal on all three comparators.
Dead-time logic	Three hand-built dead-time circuits failed (one-sided delay, boolean/double type mismatch, wires drawn as two separate lines instead of a branch).	The hand-built logic was abandoned. The final model uses ideal switches without dead time, which is a stated limitation.
Wrong sign in the dq voltage references	Growing modulation signal. The earlier fix made the PI correction add in the voltage references.	In the restart the original subtracting convention was kept and the DC link held. The earlier change appears to have introduced positive feedback for a grid current measured into the converter.
No limits or anti-windup	Integrators wound up during the start-up transient.	Output limits and clamping were added to the earlier model; the final model avoids the start-up transient with a precharged DC link and a ramped reference.
Modulator scaling and overmodulation	Gate signals were stuck high for long periods because the modulation command exceeded the carrier range.	The scaling is now computed from the DC-link voltage: Kmod = 1000/(Vdc/2).
Unstable current-loop gains	Checked against the real LCL plant, the gains from the original method had a phase margin of about −94°.	The crossover frequency is now min(Fsw/10, Fres/2.5), roughly 1 kHz, giving a stable loop in the switching-level check.
Persistent oscillation (unresolved)	The DC link did not reach its target and the current oscillated far above rated value. The suspected cause, the current-measurement point, was never tested.	Not resolved in that model; the project was restarted from the original model.

Table C1: Phase 1 difficulties and solutions.

C.2 Phase 2: restart from the original model

Difficulty	What was observed / cause	How it was solved
SOC not increasing	The repaired model never produced a rising SOC, which is the project requirement.	Restarted from the original model and changed only values, then added the reference ramp in a separate file.
Hard-coded numbers	Grid voltage, Cdc, inductor and capacitor values, current references, limits and gains were fixed for 400 V / 200 kW.	Replaced them with workspace variables calculated by charger_design.m from the two inputs.
PWM scaling of 1	The unity scaling only worked because the integrators compensated, and the effect changes with the DC-link voltage.	Set the scaling Kmod = 1000/(Vdc/2).
DC link collapsing in a pre-check	In the switching-level Python check a 50 Hz voltage loop let the DC link collapse under load steps.	Raised the voltage-loop crossover to 150 Hz; the DC link then held at four operating points in that check.
DC-link overshoot at CC to CV	Without the ramp the DC link jumped to about 1000 V at 0.4 s.	Added a start-up ramp and a rate limiter in the RAMP model; the DC link then stayed around 700–750 V at that moment.
Start-up spike (open)	The DC link and grid current spike for a few milliseconds at t = 0.	Not removed. Delaying the PWM enable until the PLL has locked is the proposed fix.
Low-voltage ripple (open)	At 220 V / 60 kW the DC-link ripple is about 25 % peak-to-peak and the grid current looks high.	Not investigated; listed as a limitation.

Table C2: Phase 2 difficulties and solutions.

C.3 Lessons learned

- A design report does not guarantee that the simulation file implements it; each value in the model has to be traced back to the design.
- When a change that should matter has no visible effect, check the process first: stale workspace, a model already open under the same name, or a clipped scope axis.
- Check controller gains against the real plant, not only the simplified one used to derive them.
- Restarting from a known-working model and changing one thing at a time was faster than repairing a model with many interacting changes.